

The Elasticity of Home Purchases to Financing Costs: Evidence from Recent GSE Pricing Changes

You Suk Kim
Federal Reserve Board

Feng Liu
CFPB

David Zhang
Rice University

June 19, 2026

Abstract

How is homebuying affected by relative changes in the price of mortgage credit? We study it using a natural experiment where the government-sponsored enterprises' (GSEs) guarantee fee (g-fee) pricing changed differently across borrower groups. We find large elasticities of home purchases to changes in fee-adjusted GSE interest rate, particularly in tight housing markets. These results suggest that interest rate subsidy programs can meaningfully induce demand in tight market conditions. Our elasticities also discipline the extensive-margin elasticity in macroeconomic models of housing and mortgage credit. We use a structural model calibrated to our estimates to evaluate alternative g-fee proposals.

Keywords: Government Sponsored Enterprises (GSEs), Securitization, Redistribution, US Mortgage Market.

JEL classification: G18, G21, G51

* The views expressed in this paper are those of the authors and are not the official positions of the Consumer Financial Protection Bureau, Federal Reserve Board or the Federal Reserve System, or the United States. We thank Natee Amornsiripanitch, Dragana Cvijanović, Andrea Eisfeldt, Rohan Ganduri, Ed Glaeser, John Griffin, Erica Jiang, Marius King, George Krivorotov, Sam Krueger, Jonathan Parker, Matthew Plosser, Tianyue Ruan, Johan Sulaeman, Sheridan Titman, Tiffany Tsai, Stijn Van Nieuwerburgh, Zhaogang Song, Laura Starks, James Vickery, Nancy Wallace, Luke Watson, Li Wei, and Dayin Zhang and seminar and conference participants at the National University of Singapore, Federal Reserve Bank of Philadelphia, the 2024 GT–FRBA Household Finance Conference, 2025 Interagency Risk Quantification Forum, 2025 Consumer Finance Round Robin, 2025 Texas Finance Festival, 2025 AREUEA National Meeting, 2025 NBER Summer Institute Real Estate, 2026 MFA, and 2026 NUS-Baruch Real Estate Conference for their valuable comments which greatly improved the paper. All errors are our own.

1 Introduction

The elasticity of home purchases to financing costs is important for evaluating the impact of policies influencing the price of mortgage credit, such as the GSE guarantee fee (g-fee) policy as well as other interest rate policies such as the FHA mortgage insurance premium, interest rate subsidies offered to veterans and the workforce of some universities. Beyond its magnitude, understanding the channels through which financing costs affect home purchases is also important for validating widely used models of the housing market such as the user-cost models as well as modern macro-housing models of how mortgage credit affects house prices. Specifically, these models assume households respond to financing costs when making home purchase decisions, which implies a sizable elasticity beyond the underwriting constraints of debt-to-income (DTI) limits.¹

Yet empirical evidence remains limited. [Bhutta and Ringo \(2021\)](#) estimate the elasticity using a natural experiment, but with a sample restricted to financially constrained borrowers. Among that group, they also find that financing cost effects are entirely mediated by DTI constraints, which if broadly applied would imply that both user-cost models and modern macro-housing models of mortgage credit require modification. [DeFusco and Paciorek \(2017\)](#) estimates an intensive margin elasticity of home purchases to interest rates, but is silent on the extensive margin of home purchases. As a result, it remains unknown how the effect of financing costs on home purchases operate among less constrained borrowers, and how much they vary across market environments.

We provide one of the first natural experiment evidence on how home purchases respond to financing costs in a broad population of U.S. households, including across market tightness conditions and by borrower counterfactual DTI. Using a 2023 adjustment of the government-sponsored enterprise (GSE) guarantee fees that changed mortgage pricing sharply across credit score buckets, we estimate large semi-elasticities of 21–25% in home purchases per 50 bps

¹Specifically, the user cost model of ([Poterba, 1992](#); [Glaeser et al., 2012](#)) require a large elasticity of home purchases to mortgage interest rates among marginal homeowners to enforce a no-arbitrage condition between owning and renting. The modern macro-economic model of housing markets and mortgage credit of [Greenwald and Guren \(2025\)](#) imply roughly half of the elasticity of home purchases to financing costs from non-DTI channels, and another half from DTI channels, as shown in Appendix J and as is consistent with our estimates.

differential decrease in fee-adjusted GSE interest rate across borrowers with similar credit scores, with approximately half of the effect coming from DTI constraints. The large effects are driven by tight housing markets, with the semi-elasticity more than doubling from the loosest to the tightest quartile. We use a model to interpret these estimates and explore counterfactual g-fee designs. Our estimates support modern macro-economic models of housing with home buyer responsiveness to financing costs and house supply constraints, and inform ongoing policy debates over GSE pricing and capitalization. Our results also suggest that interest rate subsidies can have a sizable impact on the home purchases, particularly in tight housing markets.

For identification, we exploit the May 2023 restructuring of the GSE Loan-Level Price Adjustment (LLPA), which changed g-fees discontinuously across credit score and loan-to-value ratio (LTV) grids. We use a stacked difference-in-differences (DID) design (Cengiz et al., 2019) comparing borrowers just above versus just below five credit score cutoffs (640, 680, 720, 760, and 780), where g-fee changes were particularly large. After confirming no credit score manipulation or pre-trends, we estimate how relative home purchase volumes respond to g-fee differentials.² Because our design compares borrowers just above versus just below each cutoff, our estimate captures relative rather than absolute elasticities. Any spillover between similar borrowers—such as if they search for homes in the same market with limited inventory—can affect the magnitude of our estimate. Thus, we use a structural model to account for such spillover and evaluate aggregate policy effects, which we discuss later.

Our estimates show that home purchase volume is strongly sensitive to g-fee changes, even at narrow credit score cutoffs, with those receiving a reduction in g-fees increasing their home purchases relative to those experiencing an increase or a smaller reduction. Overall, a 100 basis point (bps) decline in g-fees, which by our estimates leads to a 28–33 bps decline in fee-adjusted GSE interest rate, leads to a 13.9% increase in relative home purchase originations compared to borrowers who experienced less of a decline. This translates into a 21–25% increase in home purchases per 50 bps decrease in fee-adjusted GSE interest rate. We further show that

²We focus on credit score rather than LTV cutoffs because LTVs are endogenous for liquidity-unconstrained households who can adjust down payments.

this responsiveness is driven by both DTI-constrained and DTI-unconstrained borrowers: a subsample of DTI-unconstrained households exhibit smaller, but still sizeable, home purchase elasticities that are half as high as the elasticities for DTI-constrained households. This result is some of the first evidence of a negative elasticity of home purchases to financing costs outside of its mechanical effect through the DTI constraint.

We further investigate how our elasticities vary by housing market tightness, measured by county-level days on market (DOM) for homes listed for sale. We find that the semi-elasticity more than doubles as we move from the loosest to the tightest quartile of housing markets, while other confounders do not vary strongly across the quartiles. This pattern further points to a supply constraint channel: in tight markets where inventory is limited, differences in financing costs can have larger effects on relative volumes by shifting which buyers succeed in purchasing homes. This result supports a modeling approach that incorporates both buyer responsiveness to financing costs and supply constraints—such as the assignment models of [Landvoigt et al. \(2015\)](#) and [Hacamo \(Forthcoming\)](#), and macroeconomic models with supply constraints as in [Greenwald \(2018\)](#) and [Greenwald and Guren \(2025\)](#)—and can help their parameterization.³ It also suggests that if market tightness on average relaxes from 2023 levels, the elasticity of home purchases to financing costs may be reduced.

We now benchmark our estimate to the limited existing estimates of the home purchase elasticity to borrowing costs in the literature. Our estimate is 1.5 to 1.8 times larger than the 14% relative increase in home purchases by low credit score, FHA-likely borrowers in response to a 50 bps decline in FHA interest rates found in [Bhutta and Ringo \(2021\)](#), who found the effect to be entirely driven by DTI constraints. It is also approximately 2 times larger than the 11.8% effects per 50 bps aggregate increase in interest rate in [Hacamo \(Forthcoming\)](#) for middle-priced houses. These differences are plausibly explained by the tightness of housing markets in our sample period—which we show amplifies relative elasticities—and by the different policies and populations studied. Importantly, consistent with [Bhutta and Ringo \(2021\)](#), we find that among highly leveraged households in our sample, the elasticity is entirely mediated

³Appendix Section J derives the home purchase elasticities from DTI vs non-DTI constrained borrowers in [Greenwald and Guren \(2025\)](#), and finds that the share is similar to our estimates.

by DTI constraints, which both validates our identification approach and highlights that our broader sample reveals substantial non-DTI responsiveness absent in the financially constrained population.

We use our reduced-form estimates to discipline a structural housing search model, which helps address the missing intercept problem ([Wolf, 2023](#)) in our relative elasticity estimates as well as allowing us to extrapolate them to policy changes other than one from May 2023. Our model incorporates frictional housing search that fits housing search moments well, and naturally incorporates how the limited home inventory amplifies the relative home purchase elasticity. The model features endogenous buyer and seller entry and loan-type choice through a logit demand system, heterogeneous buyer valuations that depend on their reference mortgage interest rate, and a subsequent continuous-time search and matching game in which buyers and sellers are matched and negotiate over price. We discipline the model with a rich set of in-sample housing search moments, including buyers' weeks searched before purchase, sellers' days on market, number of offers before success, average house price, idiosyncratic house price dispersion, and buyer pre-approval to home purchase success probability, all of which the model fits with reasonable parameters.

The out-of-sample aggregate elasticities implied by our model, up to 16.4% per 100 bps decline in interest rates depending on our supply side assumptions, are significantly lower than our reduced-form relative elasticities used to fit the model. This reflects general equilibrium effects: when home inventory is limited, policies that differentially affect borrowers are likely to reallocate who purchases homes, an effect absent when uniform policies affect all borrowers. Out of sample, the model aligns well with untargeted elasticity estimates in the literature, including the intensive-margin elasticity of [DeFusco and Paciorek \(2017\)](#), the FHA extensive-margin elasticity of [Bhutta and Ringo \(2021\)](#), and the aggregate interest rate elasticity estimated in [Hacamo \(Forthcoming\)](#), lending credibility to its quantitative implications for policy counterfactuals.

We use the model to evaluate three counterfactual exercises, each under both fixed seller supply and free entry of sellers, bracketing the range of supply-side responses across U.S. housing markets. Our first counterfactual examines a uniform 25 bps increase in upfront g-

fees across all FICO-LTV cells, a policy often contemplated to improve GSE capitalization as the GSEs prepare to exit conservatorship. Evaluating this policy requires demand elasticity estimates, as the impact on GSE capitalization depends critically on borrowers responses. We find that GSE volume declines by 3.1–5.9%, with total home purchases declining by up to 2.6% when seller supply responds endogenously. There is substantial heterogeneity across credit scores: high-credit-score borrowers reduce purchases while low-credit-score borrowers increase theirs, as the latter rely more heavily on FHA loans that become relatively more attractive with higher g-fees. Despite volume declines, GSE profits rise by \$1.26–1.39 billion annually, approximately 1% of the GSEs’ combined equity as of end-2023, as higher markups more than offset volume reduction. These results indicate that marginal g-fee increases remain profitable despite large demand elasticities, and that GSE pricing decisions have important spillover effects on the broader mortgage market beyond the GSE segment.

Our second counterfactual examines the same uniform 25 basis point g-fee increase in a looser housing market condition (higher DOM), recalibrating seller entry until average days on market reaches 64 days, the 2019 pre-pandemic average. We find that market tightness reshapes the incidence of the policy but not its overall impact on the GSEs. In general, the competition channel is reduced, and so the increase in home purchases by low-credit-score borrowers is reduced under this counterfactual. The effect on GSE profits, however, is between \$1.24 and \$1.40 billion annually, nearly identical to the baseline market, indicating that the GSEs’ ability to improve capitalization through g-fee increases remains even under higher DOM conditions.

Our third counterfactual simulates the effects of the May 2023 LLPA reform itself. We find that the reform, which brought g-fees closer to the model-implied estimates of credit risk from (Capponi et al., 2026), generated substantial reallocation across credit scores but minimal aggregate effects. Middle credit score borrowers (680–779) reduced home purchases by 1–4% as their g-fees increased, while borrowers with credit scores below 680 or at least 780 increased purchases as their g-fees declined. Accounting for these demand responses, the reform increased aggregate GSE profits by \$0.24 billion annually (0.2% of GSE equity as of end-2023), as higher revenues from middle credit score borrowers more than offset lower revenues at the tails. Furthermore, the reform generated only small effects on home purchases along

the income dimension, consistent with the weak correlation between credit scores and income (Beer et al., 2018). These results suggest that the policy improved alignment between pricing and risk, modestly enhanced GSE capitalization, and had limited distributional consequences along the income dimension, which contrasts with the significant media attention paid to the policy change.

The remainder of the paper is organized as follows. Section 2 provides a literature review. Section 3 provides institutional background on GSE pricing and the May 2023 LLPA reform. Section 4 describes our data. Section 6 examines the pass-through of g-fee changes to borrowing costs. Section 7 presents our main reduced-form estimates of the effects of g-fees on relative home purchases. Section 8 develops and calibrates the structural model of housing search. Section 9 presents counterfactual exercises, and Section 10 concludes.

2 Literature Review

We contribute to the literature that measures the elasticity of mortgage demand to interest rates. This literature includes DeFusco and Paciorek (2017), Bhutta and Ringo (2021), and Hacamo (Forthcoming). We differ from the previous literature in three ways. First, we estimate extensive-margin home purchase elasticities among a broad set of borrowers who take out GSE mortgages. Second, we use a natural experiment that differentially shifted financing costs for borrowers at sharp credit score cutoffs. Third, we identify a negative elasticity of home purchases to financing costs outside of DTI constraints, which is particularly large in tight housing markets. Our results thus support modeling approaches that incorporate buyer responsiveness to interest rates and supply restraints, including the assignment models of Landvoigt et al. (2015) and Hacamo (Forthcoming) and macro-housing models of Greenwald (2018) and Greenwald and Guren (2025), and can help the parameterization of future models of this style.

Furthermore, our paper contributes to the large literature on housing search by developing a model with heterogeneous buyer financing costs. This literature is large, and includes Wheaton (1990), Krainer (2001), Albrecht et al. (2007), Novy-Marx (2009), Piazzesi and Schneider

(2009), [Genesove and Han \(2012\)](#), [Ngai and Tenreyro \(2014\)](#), [Head et al. \(2014\)](#), [Gabrovski and Ortego-Marti \(2019\)](#), [Ouazad and Ranci re \(2019\)](#), [Piazzesi et al. \(2020\)](#), [Anenberg and Bayer \(2020\)](#), [Guren and McQuade \(2020\)](#), [Gabrovski and Ortego-Marti \(2021\)](#), [Sagi \(2021\)](#), and [Gerardi et al. \(2024\)](#). Within this literature, our method of tractably endogenizing buyer and seller entry as well as endogenizing buyer financing costs is new and may be usable in other settings. Our model of housing search allows us to naturally incorporate the effect of days on market (DOM) on the elasticity of home purchases to financing costs while matching standard housing search moments.

A first stage for our home purchase elasticities is the extent to which the g-fees are pass-through to borrowers. Several contemporaneous working papers examine this issue. [Amorn-siripanitch and Ricks \(2024\)](#) examines the pass-through of these g-fees to borrowing costs over the same period, and find symmetric pass-through across g-fee increases and decreases that may or may not be considered complete depending on the conversion factor used between interest rate and fees. In addition, they find elevated rejection rates in response to g-fee increases among small dollar mortgages. In contrast to the symmetric pass-through result, [Kalda et al. \(2025\)](#) finds some asymmetry in pass-through by the direction of g-fee change. Both of these papers examine pass-through in loan-level CS by LTV cells for all affected loans, whereas we use a stacked DID specification at a more limited set CS cut-offs to be consistent with the specification in our second stage. [Adelino and Ahsin \(2025\)](#) studies an earlier g-fee change and finds complete pass-through. Whether the first stage pass-through is complete or not is not important for our results, as we only use our first stage estimates to interpret our second stage. Interestingly, we find some pass-through of g-fees to non-agency conforming mortgages among non-bank lenders, which is novel to this literature, and is potentially indicative of the pricing of aggregators who purchase loans from non-banks and subsequently keep some of them in portfolio ([Benson et al., 2024](#); [Zheng, 2024](#)).

Finally, our paper contributes to policy discussions regarding the role of GSEs, which have a large impact on the market through their credit guarantee ([Lucas and McDonald, 2010](#); [Jeske et al., 2013](#); [Elenev et al., 2016](#); [Gete and Zecchetto, 2018](#); [Grundl and Kim, 2021](#); [Gete et al., 2023](#); [Bosshardt et al., 2024b](#); [Adelino et al., 2025](#)) and the liquidity of their mortgage-backed

securities ([Vickery and Wright, 2013](#); [Huh and Kim, 2022](#)). Our reduced form estimates and model counterfactuals speak to the effects of GSE pricing policies on home purchases and whether the GSEs can increase their fees to improve their capitalization.

3 Background

The GSEs charge two types of g-fees to lenders selling loans to them: ongoing g-fees and upfront g-fees. The ongoing g-fees are monthly charges and do not vary by loan-level credit characteristics. The upfront g-fees is a one-time fee per loan paid at the time of the delivery through risk-based pricing of the loan level price adjustments (LLPAs), which primarily vary by credit score and LTV.⁴ This risk-based pricing contrasts with the Federal Housing Administration (FHA) and Department of Veteran Affairs (VA) loans, which a flat rate/premium regardless of loan characteristics.

Figure 1(a) shows that the upfront fee (in percent of the loan amount) generally increases for loans with higher credit risk (lower credit scores and higher LTVs). The LLPAs had remained largely stable from the Great Financial Crisis until January 2023, when the GSEs announced major changes taking effect May 1, 2023.⁵ Importantly, first-time home buyers with low to moderate incomes (LMI) were exempt from the LLPA starting in December 2022 and are unaffected by the May 2023 LLPA change.

Figure 1(b) shows changes in upfront g-fees by credit score and LTV for home-purchase mortgages. Red cells indicate LLPA increases; green cells indicate decreases. The changes vary sharply across discrete bins: a borrower with credit score 679 and LTV 80 saw an 87.5 bps decline, while an otherwise identical borrower with credit score 680 saw no change. For LTVs above 60, all borrowers just below the 680 cutoff received more favorable changes than those just above it, regardless of their LTV choice.

Figure 1(c) presents the average change in g-fee by credit score bin, weighted by the LTV distribution of GSE loans originated before March 2023. The differences in changes are largest

⁴Fannie Mae refers to upfront fees as “loan level price adjustments,” while Freddie Mac refers to them as “credit fees in price.” Because the industry typically uses the term LLPAs, in this paper we adopt the Fannie Mae term.

⁵See the directive [HERE](#) and an overview of the changes [HERE](#).

at credit score cutoffs of 640, 680, 720, 760, and 780. Our research design exploits these differential changes to identify effects on home purchase originations and other outcomes. Our approach yields tight overall standard errors and dynamic plots that suggest limited pre-trends. Section 5 discusses our identification strategy in more detail.

4 Data

Our main analysis uses the confidential HMDA data from 2022 to 2023, which include loan origination dates, credit scores, property values, and loan amounts. This information allows us to determine the amount of the upfront g-fee and its change for each loan using the LLPA matrix shown in Figure 1(b).

We focus on conventional, FHA, and VA home-purchase loans, excluding refinances and USDA loans.⁶ We exclude borrowers with low-to-moderate incomes (LMI) because first-time LMI home buyers were exempt from upfront g-fees starting December 2022. Since HMDA does not identify first-time buyers, we exclude all LMI borrowers.⁷ Our sample therefore focuses on relatively less constrained households with household income above the area median household income, which represents a majority of home buyers and which further distinguishes our study from the sample of FHA borrowers studied in the earlier literature (Bhutta and Ringo, 2021).

We identify GSE loans using HMDA’s “purchaser type” field. Conventional loans not sold to the GSEs are classified as “non-agency.” We group FHA and VA loans together as both have similar credit profiles and receive Ginnie Mae guarantees.

The “purchaser type” field has two known issues. First, small reporters often sell through aggregators rather than directly to GSEs. We address this by using Universal Loan IDs to track loans from originator to final purchaser, identifying an additional 17 percent of GSE loans. Second, loans originated late in the year may not be sold before year-end reporting, appearing as “not sold” despite eventual GSE purchase.⁸ We therefore drop December originations from

⁶Refinance borrowers may time applications strategically to secure favorable LLPA changes, and refinance activity was minimal during this period due to the interest rate environment.

⁷LMI is defined as income below the area median household income.

⁸The median time from origination to sale is 16 days (mean: 32 days; 95th percentile: 56 days).

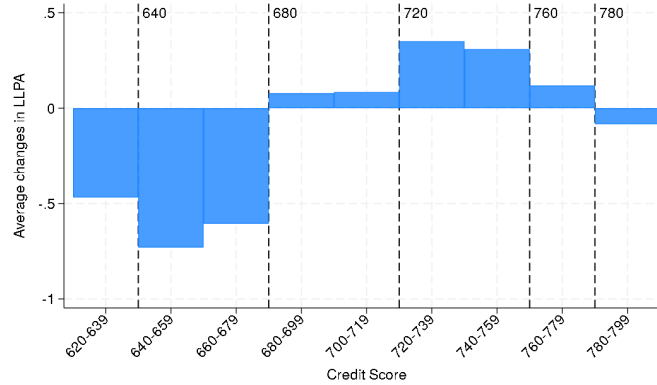
Figure 1: LLPA Matrix and Its Change in May 2023

Representative Credit Score	LTV Range								
	Applicable for all mortgages with terms greater than 15 years								
	≤ 60.00%	60.01 – 70.00%	70.01 – 75.00%	75.01 – 80.00%	80.01 – 85.00%	85.01 – 90.00%	90.01 – 95.00%	95.01 – 97.00%	>97.00%
≥ 740	0.000%	0.250%	0.250%	0.500%	0.250%	0.250%	0.250%	0.750%	0.750%
720 – 739	0.000%	0.250%	0.500%	0.750%	0.500%	0.500%	0.500%	1.000%	1.000%
700 – 719	0.000%	0.500%	1.000%	1.250%	1.000%	1.000%	1.000%	1.500%	1.500%
680 – 699	0.000%	0.500%	1.250%	1.750%	1.500%	1.250%	1.250%	1.500%	1.500%
660 – 679	0.000%	1.000%	2.250%	2.750%	2.750%	2.250%	2.250%	2.250%	2.250%
640 – 659	0.500%	1.250%	2.750%	3.000%	3.250%	2.750%	2.750%	2.750%	2.750%
620 – 639	0.500%	1.500%	3.000%	3.000%	3.250%	3.250%	3.250%	3.500%	3.500%
< 620 ¹	0.500%	1.500%	3.000%	3.000%	3.250%	3.250%	3.250%	3.750%	3.750%

(a) LLPA Matrix for 2022

Credit Score	LTV Range								
	Applicable for all loans with terms greater than 15 years								
	≤ 30.00%	30.01 – 60.00%	60.01 – 70.00%	70.01 – 75.00%	75.01 – 80.00%	80.01 – 85.00%	85.01 – 90.00%	90.01 – 95.00%	>95.00%
≥ 780	0.000%	0.000%	-0.250%	-0.250%	-0.125%	0.125%	0.000%	0.000%	-0.625%
760 – 779	0.000%	0.000%	-0.250%	0.000%	0.125%	0.375%	0.250%	0.250%	-0.500%
740 – 759	0.000%	0.000%	-0.125%	0.125%	0.375%	0.750%	0.500%	0.375%	-0.250%
720 – 739	0.000%	0.000%	0.000%	0.250%	0.500%	0.750%	0.500%	0.375%	-0.250%
700 – 719	0.000%	0.000%	-0.125%	-0.125%	0.125%	0.500%	0.250%	0.125%	-0.625%
680 – 699	0.000%	0.000%	0.125%	-0.125%	0.000%	0.375%	0.250%	0.125%	-0.375%
660 – 679	0.000%	0.000%	-0.250%	-0.875%	-0.875%	-0.625%	-0.500%	-0.625%	-1.000%
640 – 659	-0.500%	-0.500%	-0.125%	-1.250%	-0.750%	-0.750%	-0.750%	-0.875%	-1.250%
620 – 639	-0.500%	-0.375%	0.000%	-0.875%	-0.250%	-0.375%	-0.625%	-1.000%	-1.750%
≤ 619 ¹	-0.500%	-0.375%	0.000%	-0.875%	-0.250%	-0.375%	-0.625%	-1.000%	-2.000%

(b) LLPA Changes in May 2023



(c) Average LLPA Changes by Credit Score

Note: Figure 1(a) shows the level of the LLPA for loans sold to the GSEs in 2022. Figure 1(b) shows the changes in the LLPA for home purchase loans effective in May 2023. The values are in percentages of the loan amount. Green cells represent decreases in g-fees, while red cells represent increases. Figure 1(c) shows the average LLPA by credit score bin, weighted by the LTV distribution of GSE loans originated before March 2023.

analyses where GSE status matters for the outcome variable.

4.1 Summary Statistics

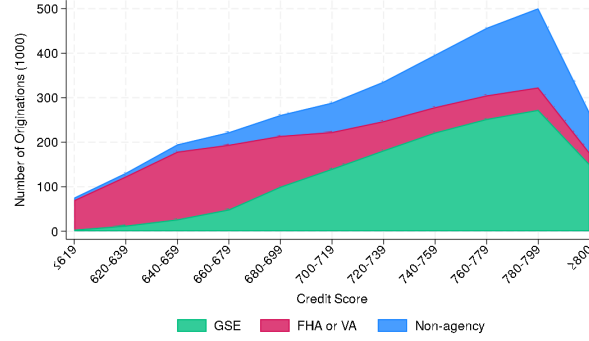
Table 1 presents the summary statistics for our sample period separated by three loan types that we consider: GSE loans, FHA or VA loans, and non-agency loans. On average FHA and VA loans have much higher LTVs and lower credit scores than GSE and non-agency loans. The average property values are much higher for GSE and non-agency loans than for FHA or VA loans, while origination loan sizes are not much different across loan type, likely due to the offsetting effects of LTVs. DTIs are lower for GSEs and non-agency loans than for FHA or VA loans, suggesting that households with lower incomes are less likely to choose GSE or non-agency loans. The average mortgage rate is about 5.6 percent in our sample period. Lastly, borrowers in the sample pay about \$3,000 for points and origination fees net of lender credits, which is about 90 bps of the average loan origination balance.

We further illustrate how loan and borrower characteristics may vary by credit scores in Figure 2. Figure 2(a) shows the number of home-purchase loan originations in 2022 for each of the three loan types. Figures 2(b) and 2(c) plot the average LTV and the average household income relative to the median family income of the relevant Metropolitan Statistical Area or Metropolitan Division by loan type and credit score, respectively. Together these three figures illustrate a large degree of heterogeneity in loan characteristics among households with different credit scores. This large heterogeneity implies that the effects of changes in g-fee pricing will be heterogeneous among households with different credit scores and incomes.

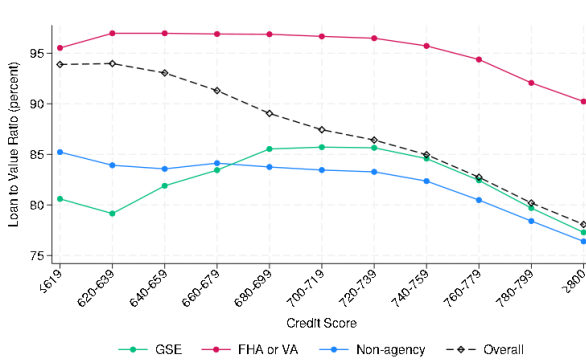
Figure 2(a) shows that GSE originations account for significant shares of total loan originations for households with credit scores at least 680, suggesting that lower LLPA for higher credit scores make GSE loans a cheaper option for such households than FHA or VA loans. Non-agency loans also account for higher shares for higher-score households likely because of their lower credit risks. Figure 2(b) shows that the average LTV of GSE loans for households with credit scores at least 680 is slightly higher than that for households with credit scores below 680, but the average LTV of GSE loans always remains much lower than the average LTV for FHA or VA loans because the LLPA is still higher for higher LTVs even for higher

credit scores. Figure 2(c) shows that, for most credit score bins, households with GSE or non-agency loans generally have higher incomes relative to the median family income of the area where their homes are located.

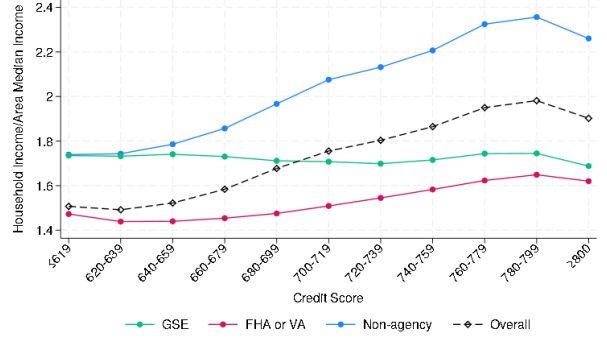
Figure 2: Loan Characteristics by Credit Score, Home Purchase Loans



(a) Number of Originations



(b) Average LTV



(c) Average Ratio of Household Income to Area Median Family Income

Note: Panel (a) shows the number of home-purchase loan originations for each loan type by credit score. Panel (b) shows the average LTVs of each loan type by credit score. The sample is loans originated in 2022, excluding December.

5 Empirical approach

5.1 Regression specifications for loan-level outcomes

We estimate the effects of g-fee changes using a stacked DID design that exploits discontinuities in the May 2023 LLPA restructuring around the four major credit score cutoffs. The baseline

Table 1: Summary statistics by loan type

	GSE	Non-Agency	FHA/VA	All
Credit Score	752 (42)	749 (45)	689 (55)	731 (56)
LTV	81.7 (15.6)	81.98 (16.0)	97.0 (8.2)	86.8 (15.4)
DTI	37.7 (8.9)	36.6 (9.1)	44.4 (8.9)	39.7 (9.5)
Loan Amount (\$000s)	344.1 (165.7)	338.7 (187.5)	316.0 (137.7)	334.0 (161.7)
Property Value (\$000s)	436.7 (228.9)	435.4 (274.0)	330.4 (148.7)	401.5 (220.8)
Interest Rate (%)	5.647 (1.227)	5.793 (1.414)	5.560 (1.205)	5.643 (1.256)
Net Origination Charges (\$)	3,084 (3,316)	2,849 (3,443)	2,827 (3,376)	2,960 (3,360)
Share of Loans (%)	50.3	16.9	32.9	100

Note: The sample consists of loans originated in 2022 and 2023, excluding Decembers, limited to closed-end, non-reverse mortgages, 30-year fixed rate, non-IO, home-purchase mortgages secured by site-built, one-unit, first-lien, principal residence. Standard deviations are in parentheses.

specification for loan-level outcomes (e.g., interest rate) is:

$$y_{i,c,k,t} = \beta \times \mathbf{1}[t > \text{Feb 2023}] \times \mathbb{E}[-\Delta LLPA_c] + \xi_{C_g,k,t} + \xi_c + \epsilon_i, \quad (1)$$

where $y_{i,c,k,t}$ is the outcome variable (e.g., interest rate) for individual loan i originated by a borrower with credit score c , income bin k , in year-month t . The treatment variable $\mathbb{E}[-\Delta LLPA_c]$ represents the expected LLPA decline for the LLPA credit score bin containing credit score c , weighted by the pre-treatment LTV distribution within that bin. For example, a borrower with credit score 679 is assigned the expected LLPA change for the 660–679 LLPA bin, calculated by averaging the LLPA changes across LTVs using the pre-treatment LTV distribution of borrowers in that bin. We focus on differences in how much g-fees changed across borrowers with different credit scores rather than different LTVs because LTVs can be endogenous. The unit of the treatment variable is percentage points, so our coefficient of interest β measures the effect of a 100 bps LLPA decline on the outcome variable. We assume

the new LLPA schedule became effective for loans originated in March 2023 or later, as it typically takes up to two months before securitization into MBS issued by the GSEs.

We construct five distinct estimation windows around the major LLPA credit score cutoffs ($g \in \{640, 680, 720, 760, 780\}$). Each cutoff group C_g encompasses a 20-point credit score band ranging from $g - 10$ to $g + 9$. For example, the group C_{680} centered at 680 includes all credit scores from 670 to 689. This bandwidth exactly matches the size of the standard LLPA credit score bins, aligning with the GSEs' practice of grouping borrowers with similar credit profiles into homogeneous pricing buckets. Our regression just considers a different way to slice the credit score distribution from the one considered in the LLPA pricing. Any credit scores that fall outside these five localized ranges are dropped from the estimation sample.

The specification includes two sets of fixed effects. First, by stacking the five cutoff groups, we include the high-dimensional fixed effect $\xi_{C_g, k, t}$, which sweeps out any common trends affecting borrowers within the same cutoff group C_g and income bin k in a given month. Consequently, identification is driven strictly by comparing borrowers with similar credit risk who fall on opposite sides of a specific credit score cutoff within the same group C_g : those with credit scores from $g - 10$ to $g - 1$ versus those from g to $g + 9$, where the policy-induced LLPA changes differed sharply. We consider 4 income bins, based on the distribution of a mortgage borrower's income relative to the area median income (AMI): (i) < 33 th percentile, (ii) $[33$ th percentile, 66 th percentile), (iii) $[66$ th percentile, 90 th percentile), (iv) ≥ 90 th percentile. The fourth income bin captures mainly borrowers with very large incomes relative to their corresponding AMIs (ranging from $3 \times \text{AMI}$ to more than $10 \times \text{AMI}$), who may behave differently from other income groups.

Second, the credit score fixed effects ξ_c control for time-invariant differences in borrower characteristics and loan outcomes across the distinct credit scores. When examining certain outcomes, we also include additional controls such as LTV or lender fixed effects when they are likely to affect the outcome variable. Standard errors are clustered by credit score because our treatment variable is determined by credit score.

5.2 Poisson regressions for number of loan originations

Because HMDA does not provide information about would-be homebuyers who did not purchase homes, we use the number of home-purchase loan originations to examine the effects of the LLPA on home purchases. Given that loan volume is measured in counts, we estimate a Poisson regression (Cohn et al., 2022):

$$\mathbb{E}[y_{c,k,t}] = \exp(\beta \times \mathbf{1}[t > \text{Feb 2023}] \times \mathbb{E}[-\Delta LLPA_c] + \xi_{C_g,k,t} + \xi_c), \quad (2)$$

where $y_{c,k,t}$ is the number of loan originations for credit score c , income bin k , and year-month t . The treatment variable, fixed effects, and sample construction follow the same specification as in Equation (1).

With the Poisson specification, the coefficient of interest β captures the semi-elasticity of home purchase volumes with respect to a 100 bps LLPA decline. Crucially, because our design compares borrowers just above versus just below each credit score cutoff within the same cutoff group C_g , income bin, and time period, β measures a *relative* demand response rather than an absolute elasticity. If a housing market is tight with many prospective home buyers relative to houses available for sale, these observationally similar borrowers may compete in the same housing markets and likely bid for the same houses. Any equilibrium spillover effects between them—such as one group crowding out the other in tight housing markets when they receive more favorable financing terms—will be differenced out in our within-cutoff-group comparison. In such a case, β may capture how financing cost changes redistribute home purchases between similar borrowers. This is a version of the “missing intercept” problem. To address this limitation, we embed our reduced-form estimates into a structural housing search model in Section 8, which explicitly accounts for these equilibrium effects.

Concerns about endogenous credit scores A potential threat to our identification is the possibility that credit scores on mortgage applications are endogenous to the g-fee change. However, our identification does not require the complete absence of credit score manipulation, but only that borrowers do not change their manipulation behavior in response to the LLPA

change. For example, if homebuyers generally pay down non-mortgage debt before entering a house search to improve their credit scores—regardless of the LLPA policy—our identifying assumption would not be violated. The assumption would be violated only if borrowers systematically altered their credit-building efforts in response to the LLPA changes, becoming less (more) likely to manipulate their credit scores when the LLPA decreased (increased). If such sophisticated, policy-responsive manipulation were prevalent, we would overstate the effects of the LLPA on home-purchase originations.

A priori, it is likely very difficult for borrowers to engage in strategic manipulation in response to the LLPA change. First, the LLPA change was not uniform across credit score cutoffs. Around cutoffs such as 680 and 720, the LLPA declined more for credit scores below the cutoffs than above. Around other cutoffs such as 760 and 780, the LLPA increased more for credit scores below the cutoffs. Second, the credit score used for underwriting is the median of three credit scores from the three credit bureaus, making it challenging for borrowers to know precisely where they stand relative to specific cutoffs and to successfully shift across them.

In Appendix [A](#), we provide multiple tests for endogenous credit score responses using panel data on borrowers’ credit score evolution prior to obtaining a mortgage. We find that the likelihood of adding a co-borrower on a loan—which can change the underwriting credit score—does not respond to the LLPA change. Moreover, we find no evidence of changes in credit score manipulation patterns in response to the LLPA restructuring. These results also bolster the credibility of other studies that exploit sharp credit score cutoffs for identification, such as [Keys et al. \(2010\)](#), [Collier and Ellis \(2024\)](#), and [Collier et al. \(2024\)](#).

6 The effect of g-fees on financing costs

In this section, we examine the effect of g-fees on financing costs, which is useful for putting our home purchase elasticities in context. Because mortgage pricing is multi-dimensional, we consider not only mortgage interest rates but also the upfront fees and points paid by the borrowers directly to lenders. For the upfront price, we calculate the net origination charges, defined as fees and points paid by borrowers net of lender credits. Examples of origination

charges include origination fees, underwriting fees, application fees, as well as discount points borrowers pay to buy down the interest rate. We measure the net origination charges in percentages of the loan amount at origination.

Table 2: Effects on mortgage rates and net origination charges

	Mortgage Rate				Net Orig Charges			
	(1) GSE	(2) Non-agency conforming	(3) Jumbo	(4) FHA/VA	(5) GSE	(6) Non-agency conforming	(7) Jumbo	(8) FHA/VA
$1[t > Feb23] \times E[-\Delta LLPA]$	-0.213*** (0.013)	-0.050** (0.024)	-0.010 (0.049)	-0.024** (0.012)	-0.290*** (0.021)	-0.126*** (0.024)	-0.084 (0.051)	0.002 (0.021)
Cutoff $\times$ Income bin $\times$ Orig YM	Y	Y	Y	Y	Y	Y	Y	Y
CS FE	Y	Y	Y	Y	Y	Y	Y	Y
LLPA grid FE	Y	Y	Y	Y	Y	Y	Y	Y
Lender $\times$ County FE	Y	Y	Y	Y	Y	Y	Y	Y
N. Obs.	376,955	172,715	40,818	186,739	371,913	168,892	40,420	182,131
Adj. R^2	0.67	0.74	0.79	0.65	0.27	0.49	0.34	0.31

Note: The estimates of equation (1) for mortgage rates and net origination charges by loan type. Standard errors are clustered by credit score. Dec 2022 was excluded from the sample because of the end-of-year censoring issue in identifying whether a loan is sold to the GSEs in HMDA.

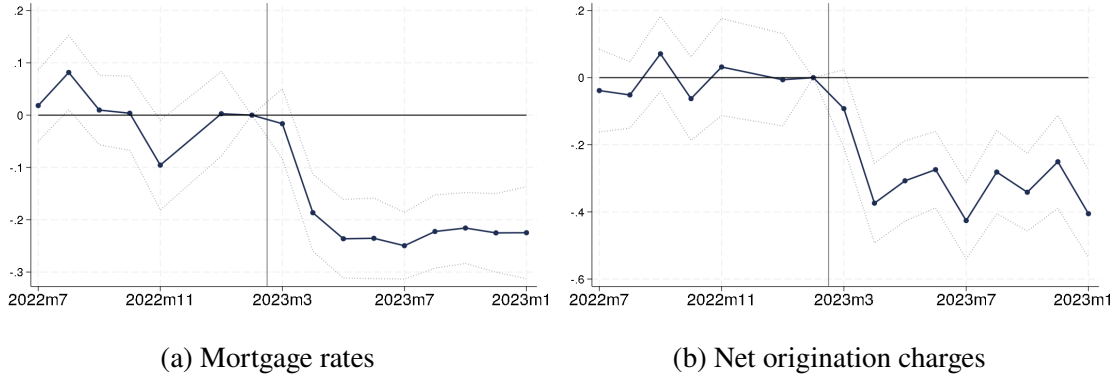
Table 2 presents the estimates of β in equation (1) for mortgage rates and net origination charges by loan type. In addition to the fixed effects in equation (1), we also include extra controls for these regressions: the fixed effects for each LLPA pricing grid (a combination of CS bin, LTV bin, and whether a loan is high-balance loans larger than the national conforming loan limit) and the lender $\times$ county fixed effects.

The estimates show that lenders passed through the LLPA change both in the form of mortgage rates and net origination charges. Columns 1 and 4 show that a 100 bps LLPA decline led to declines in mortgage rates and net origination charges for GSE loans by 21 bps and 29 bps, respectively. We use these numbers for our model.

Figure 3 presents the estimates of an event-study DID of equation (1), where we interact the treatment variable $E[-\Delta LLPA_c]$ with the month fixed effects. These results show that the estimate for GSE loans in columns (1) and (4) in Table 2 are indeed driven by changes that happened for loans originated in March 2023 or later. The effects in March 2023 are much smaller than later months because only a small fraction of loans originated in March are sold to the GSEs in May or later. Moreover, although the new LLPA was announced in January 19,

2023, we find no effects for loans originated in January or February, suggesting limited lender anticipatory behavior.

Figure 3: Event-study DIDs for borrowing costs for GSE loans



Note: coefficient estimates and their 95% confidence intervals in the event-study version of equation (1).

In Appendix 6, we provide additional details on the effects on borrowing costs. Figures A.3 and A.4 show that borrowing costs sharply diverged for credit scores just below and above each of the five credit score cutoffs in the sample, suggesting that time-varying unobserved heterogeneity across credit score points does not drive our estimates, and that the effect in Figure 3 is indeed identified by discontinuities in $E[-\Delta LLPA_c]$.

Table 2 also reports the pass-through effects on non-GSE loans. We find some spillover effects for non-agency conforming loans, which are conventional loans that are eligible for GSE securitization but not sold to GSEs. This result suggests that the option to sell these loans to the GSEs affects the pricing of these loans to some degrees, even if they are not eventually sold to the GSEs. Table A.6 in Appendix provides evidence that the lower pass-through rate for non-agency conforming loans is because the price of bank-originated non-agency conforming loans does not respond to the LLPA change. We discuss how the effects on borrowing costs differ between banks and nonbanks in more detail in Section 7.3.4. Lastly, we find limited spillover effects of the LLPA change on jumbo loans or FHA/ VA loans.

7 The effect of g-fees on relative home purchases

We now examine the effects of the LLPA change on relative home purchase volumes between borrowers on the opposite sides of the credit score cutoffs, using the Poisson regression in equation (2). Table 3 presents estimates of β . As discussed earlier, these estimates are semi-elasticities in response to a 100 bps LLPA decline for borrowers on one side of the credit score cutoff relative to another.⁹ The semi-elasticity for the total number of originations is almost 14% (column 1). Examining each loan type, the semi-elasticity for GSE loans is even higher at almost 30% (column 2), and the semi-elasticities for other loan types (columns 3–5) are much smaller, which suggests that GSE loan originations drove the effects on the total number of originations. Consistent with the spillover effects on borrowing costs of non-agency conforming loans in Table 2, the semi-elasticity for these loans (column 3) is positive. In contrast, the semi-elasticity for FHA loans (column 5) is negative, suggesting that there is a substitution between FHA/VA loans and GSE loans. This substitution effect is consistent with [Bhutta and Ringo \(2021\)](#), who also find a similar partial substitution after the 2015 cut in the FHA mortgage insurance premium.

Table 3: Effects on relative home purchases

	(1)	(2)	(3)	(4)	(5)
	Total	GSE	Non-agency conforming	Jumbo	FHA/VA
$1[t > Feb23] \times E[-\Delta LLPA]$	0.139*** (0.015)	0.296*** (0.028)	0.070** (0.032)	0.041 (0.066)	-0.073*** (0.022)
Market share (pre-policy)		0.42	0.27	0.09	0.22
Cutoff $\times$ Income bin $\times$ Orig YM	Y	Y	Y	Y	Y
CS FE	Y	Y	Y	Y	Y
N. Obs.	3,600	3,200	3,200	2,680	3,200
Pseudo R^2	0.82	0.83	0.79	0.87	0.77

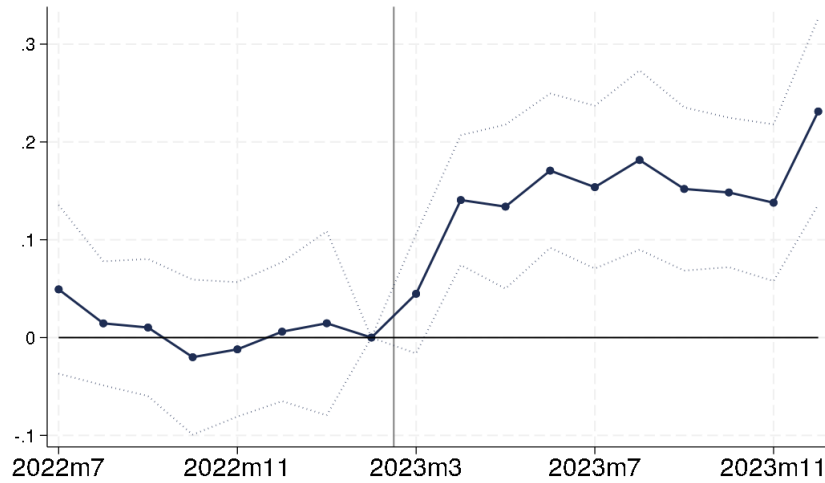
Note: this table presents Poisson regression estimates of equation (2) for the number of home-purchase originations by loan type. Standard errors are clustered by credit score.

Additional pieces of evidence support the validity of our estimates. We estimate the event-study version of equation (2) and present the estimates in Figure 4. This figure shows no

⁹The [Bhutta and Ringo \(2021\)](#) estimate is also a relative estimate by borrower credit score, and we compare to them using our model.

pre-trends, a smaller increase for March 2023, and larger increases for April and later, which is another piece of evidence that our main estimates reflect the causal effects of the change in the LLPA. A further analysis by credit score discontinuity in Appendix Figure A.5 shows that, as with financing costs, the loan volumes results are driven by sharp changes in discontinuities at the credit score cutoffs. The fact that there is no bunching at the cutoffs in Appendix Figure A.5 is another piece of evidence against credit score manipulation.

Figure 4: Event-study DIDs for the total number of home-purchase originations



Note: coefficient estimates and their 95% confidence intervals in the event-study version of equation (2).

Interpreting the estimate Because our design compares similar borrowers around credit score cutoffs, our estimate captures *relative* rather than *absolute* elasticities—how financing costs affect home purchases for one group compared to another observationally similar group, rather than the total change in aggregate purchases. In other words, by capturing a cross-sectional effect, our estimates are missing an intercept, which is a common problem in macroeconomics and which requires a model to resolve Wolf (2023). Since borrowers compete in the same market, when one group receives more favorable terms, their home purchase success may come partly at the expense of the comparison group, magnifying the relative difference we observe. The existence of such spillover effects implies that a model is required to extrapolate our home purchase elasticities to policy changes beyond the May 2023 change.

This competition channel is likely to be particularly relevant for our sample period, which

spans from July 2022 to December 2023. During this period, the housing market remained historically tight despite high interest rates, though less so than the pandemic peak. In such tight markets with multiple buyers competing for limited inventory, borrowers with better financing terms can crowd out those with worse terms. For example, a borrower with a 679 credit score qualifying at 6.95% takes out a \$450,000 30-year mortgage with a monthly payment of \$3,289. A borrower with a 680 credit score qualifying at 6.70% can maintain the same monthly payment and DTI while borrowing \$460,813, enabling them to make stronger offers and successfully purchase homes. In a tight market, a greater home purchase success probability for one borrower will reduce available inventory for other buyers.¹⁰ In Section 7.2, we provide direct empirical evidence for this mechanism: the semi-elasticity more than doubles from the loosest to the tightest quartile of housing markets. Such a mechanism suggests that individualized interest rate subsidies are likely to be more effective at increasing home purchases, but at the expense of those who do not receive the subsidies, in tight housing markets, which is novel to the literature.

Benchmarking to the literature Our estimate implies a 21 to 25% increase in home purchases in response to a 50 bps mortgage rate drop.¹¹ This magnitude is large compared to the only other estimate based on a natural experiment from [Bhutta and Ringo \(2021\)](#), who estimate that a 50 bps decrease in the 2015 FHA mortgage insurance premium—which is equivalent to a 50 bps decline in the FHA mortgage rate—increases FHA loan demand by 14%.

Multiple factors reconcile this difference. First, differences in sample composition account for part of the magnitude difference. Our estimate for highly leveraged borrowers (LTV>90) is much closer to their 14% (Section 7.1). Second, the housing market was substantially looser in 2015 compared to our historically tight 2023 treatment period. Our elasticity approaches their estimate in the loosest tightness quartile, which will be discussed in detail in Section 7.2,

¹⁰Calculated using the mortgage amortization formula: a \$450,000 loan at 6.95% requires a \$3,289 monthly payment; at 6.70%, this same payment supports a \$460,813 loan.

¹¹This conversion is based on our estimate of 13.9% for a 100 bps g-fee decrease (Column 1, Table 3) and the pass-through results in Table 2. A 100 bps g-fee decrease translates to a 28 to 33 bps reduction in the fee-adjusted interest rate, depending on the fee-to-rate conversion factor. The higher number uses a conversion factor of 0.406 as described in Section 8.4.1, where a pass-through of -0.290 percent of the loan amount in fees converts to $-0.406 \times 0.290 = -0.118$ percentage point change in rate. The lower number uses a factor of 0.218 from [Bhutta et al. \(2026\)](#). Scaling our 13.9% semi-elasticity by (50/28) to (50/33) yields the 21 to 25% range.

consistent with the competition channel.

Another factor that might drive differences in results is the magnitude of the treatment intensity. In our natural experiment, the largest change in g-fees is at the 680 cut-off, where borrowers with CS below the cut-off receives an approximately 70 basis points discount in g-fees relative to borrowers above the cut-off. In terms of interest rates, this is approximately 28 basis points in fee adjusted interest rate using the MBS conversion factor of 0.406 for our time period. This 28 basis points change is smaller than the 50 basis points change studied in [Bhutta and Ringo \(2021\)](#), but is in the same order of magnitude. The tight standard errors in Table 3 and the sharp dynamic effects in Figure 4 mitigate this as a challenge for identification.

7.1 Estimated semi-elasticities by DTI

Table 4 shows the semi-elasticity estimates for borrowers with different counterfactual DTI (“cDTI”), which is the DTI obtained by borrowers without the LLPA change. We discuss how we compute this cDTI in Appendix D. The maximum DTI allowed for GSE loans is 50%, and there is a soft DTI limit at 45% for GSE loans. Moreover, the premium for the private mortgage insurance for conventional loans with LTVs greater than 80% can be also higher for loans with DTIs above 45%. Thus, we consider borrowers with low cDTI below 40%, borrowers with cDTI close to the binding constraints, and borrowers with binding cDTI (greater than 45%).

Table 4: Effects on the relative home-purchase loan originations by counterfactual DTI

	Full sample			LTV > 90		
	(1) cDTI ≤ 40	(2) cDTI ∈ (40, 45]	(3) cDTI > 45	(4) cDTI ≤ 40	(5) cDTI ∈ (40, 45]	(6) cDTI > 45
$1[t > Feb23] \times E[-\Delta LLPA]$	0.096*** (0.021)	0.142*** (0.032)	0.182*** (0.022)	0.031 (0.032)	0.099*** (0.038)	0.129*** (0.023)
Share (pre-policy)	0.55	0.20	0.25	0.20	0.09	0.14
Cutoff × Income bin × Orig YM	Y	Y	Y	Y	Y	Y
CS FE	Y	Y	Y	Y	Y	Y
N. Obs.	3,600	3,600	3,600	3,600	3,600	3,600
Pseudo R^2	0.92	0.79	0.85	0.76	0.69	0.83

Note: this table presents Poisson regression estimates of equation (2) for relative home purchases originations. Standard errors are clustered by credit score.

Comparing the results in columns 1 to 3 of Table 4 reveals that the semi-elasticities increase in cDTI. Borrowers with low cDTI have the semi-elasticity at nearly 10% (col 1), much lower than the semi-elasticity of 18% for borrowers with binding cDTI (col 3). Thus, our estimate is in part driven by much larger elasticities among borrowers with DTI close to the maximum DTI constraint, consistent with [Bhutta and Ringo \(2021\)](#). However, the 10% semi-elasticity estimate for borrowers with low cDTIs is still sizable, although the DTI constraint is unlikely to be binding. This result suggests that even outside the DTI constraint, many borrowers are still sensitive to the size of their interest payments, as would be required in user-cost models of housing demand ([Poterba, 1992](#); [Glaeser et al., 2012](#)).

Assuming that the elasticity in households with $cDTI > 45$ is fully due to the DTI constraint, and in households with $cDTI > 40$ but $cDTI \leq 45$ is half due to the DTI constraint, the results in columns 1 to 3 of Table 4 suggest that the DTI channel and the non-DTI channel account for roughly equal shares of the full-sample effect, with DTI channels explaining approximately 43% and non-DTI channels explaining approximately 57%.¹² Interestingly, these numbers almost exactly match those implied by the model of [Greenwald and Guren \(2025\)](#), which studies how mortgage credit supply and interest rates move house prices, as described in Appendix Section J. Therefore, our study provides some of the first natural experiment evidence of a non-DTI driven elasticity of home purchases to financing costs and contributes to the validity of this style of models. These results also implies that interest rate subsidy programs by non-profits such as universities can be effective at increasing homeownership, particularly in non-tight housing markets and even for non-DTI constrained borrowers.

Columns 4 to 6 of Table 4 show the semi-elasticity estimates for a highly leveraged borrowers with LTV above 90, which is closer to FHA-likely borrowers examined by [Bhutta and Ringo \(2021\)](#). Among these households, the home purchase elasticities are concentrated among high cDTI borrowers, consistent with [Bhutta and Ringo \(2021\)](#). It is likely that highly leveraged households also face more binding DTI constraints. This result further adds to the evidence on

¹²This is assuming borrowers with $cDTI \leq 40$ have zero DTI exposure, borrowers with $40 < cDTI \leq 45$ have half exposure, and borrowers with $cDTI > 45$ have full exposure. The share with $40 < cDTI \leq 45$ is 0.20, and its semi-elasticity estimate is 0.142. The DTI component is $0.20 \times 0.5 \times 0.142 + 0.25 \times 0.182 = 0.060$, which is 43% of the semi-elasticity for the full sample (0.139 in Col 1, Table 3).

the validity of our estimates.

7.2 Estimated semi-elasticities by market tightness

We present some reduced-form evidence consistent with a competition channel, which is novel to the literature, where the elasticity of home purchases to financing costs is intensified in more competitive markets where multiple potential buyers compete for the same house. While this channel is natural in models of the housing market with supply constraints [Landvoigt et al. \(2015\)](#); [Hacamo \(Forthcoming\)](#); [Greenwald \(2018\)](#), there is limited empirical evidence on how financing costs may matter more in competitive housing markets, which matters for the effectiveness of interest rate subsidy programs. In Table 5, we interact the effect of LLPA changes on relative home purchases in column (1) of Table 3 by quartiles of market tightness. Tightness is measured as the average DOM at the county level, which we take quartiles of. We find a doubling of semi-elasticities as we go from lowest tightness quartile (row 1) to the highest tightness quartile (row 1+4). Furthermore, the GSE and non-agency shares across the four quartiles are relatively constant, between 39–45% for the former and 24–28% for the latter. We view this as evidence that the competition channel may account for between one-half and two-thirds of the semi-elasticity we estimate.

In Section 8, we build a structural model that features this competition channel. As a non-targeted moment, our calibrated model also attributes about two-thirds of our semi-elasticities of relative home purchase volumes to the competition channel, consistent with Table 5, suggesting that the competition channel emerges naturally when a standard model of housing search is enriched with heterogeneity in mortgage interest rates.

The existence of a competition channel also implies that the high interest rates in our time period, which leads to sellers “locked-in” due to having a lower interest rate on their existing home compared to the new mortgage rate they can get on the market, may have led to elevated home purchase elasticities over our sample period by reducing the supply of sellers and leading to a tighter market than otherwise ([Gerardi et al., 2026](#)). Our results on DTI points to a channel beyond lock-in driving our elasticity estimates: among potential home buyers, [Bosshardt et al. \(2024a\)](#) finds that all the missing buyers in the recent high interest

Table 5: Effects on relative home purchase origination by market tightness

	(1) 1st tightness quartile	(2) 2nd tightness quartile	(3) 3rd tightness quartile	(4) 4th tightness quartile
$1[t > Feb23] \times E[-\Delta LPA]$	0.086*** (0.021)	0.126*** (0.023)	0.156*** (0.026)	0.187*** (0.023)
Cutoff $\times$ Income bin $\times$ Orig YM	Y	Y	Y	Y
CS FE	Y	Y	Y	Y
N. Obs.	3,250	3,030	2,960	2,980
Pseudo R^2	0.90	0.89	0.90	0.91

	Median days on market (Low=Tight)		Months supply (Low=Tight)		Sold above list price (High=Tight)		Off market within 2 weeks (High=Tight)	
	(1) Low	(2) High	(3) Low	(4) High	(5) Low	(6) High	(7) Low	(8) High
$1[t > Feb23] \times E[-\Delta LPA]$	0.156*** (0.022)	0.120*** (0.015)	0.151*** (0.021)	0.124*** (0.016)	0.112*** (0.016)	0.167*** (0.021)	0.110*** (0.016)	0.169*** (0.021)
Cutoff $\times$ Income bin $\times$ Orig YM	Y	Y	Y	Y	Y	Y	Y	Y
CS FE	Y	Y	Y	Y	Y	Y	Y	Y
N. Obs.	3,040	3,350	3,020	3,340	3,320	3,050	3,280	3,090
Pseudo R^2	0.94	0.95	0.95	0.95	0.94	0.95	0.95	0.95

rate environment had counterfactual DTI that were above conventional limits, and are thus constrained by their DTI, including seller-rebuyers that were affected by “lock-in.” Thus, based on the earlier literature, all of the reduction in home buying due to the higher interest rate environment are from households who are constrained by DTI limits. Yet, we find an effect of g-fees on home buying even beyond DTI limits, suggesting an effect due to idiosyncratic changes in financing costs beyond those documented in the earlier literature.¹³ Furthermore, the financing costs changes our results are identified off of are likely too small to impact whether someone is “locked-in.” Nevertheless, mortgage lock-in may have led to tight housing market conditions, which facilitates a higher elasticity as described in this section.

Finally, we note that the existence of a competition effect also implies that the magnitude of the elasticity of home purchases to financing costs for policies other than the May 2023 g-fee change should be analyzed through the lens of a model, as different policy changes will imply different home purchase elasticities even for the same utility parameters. Nevertheless, our results generally suggests that policy that affects financing costs in relatively individualized

¹³One way to reconcile our results is by noting our results in Table 4 suggest that high DTI borrowers react *more* to financing costs compared to low DTI borrowers, but the aggregate effect of financing costs depend on general equilibrium effects which may then increase home purchases from low counterfactual DTI borrowers when given a rise in interest rates, as can be seen in Figure 1 of [Bosshardt et al. \(2024a\)](#).

ways can have substantial effects on the affected individuals' home purchases, particularly in tight housing markets.

7.3 Additional results

7.3.1 Heterogeneous effects on the number of originations by credit score and income

We also examine heterogeneity across credit score and income groups, as these patterns serve as calibration targets for the structural model (Section 8). Table A.4 shows semi-elasticities vary by borrower type. Among low credit score borrowers, higher-income households respond more strongly to the LLPA change. Lower-credit score, lower-income borrowers are more likely to lack downpayments, and FHA or VA loans may offer better pricing because of their uniform insurance pricing. Among high credit score borrowers, elasticities are relatively uniform except for very high-income borrowers (>90th percentile) who exhibit low responsiveness, likely reflecting their use of jumbo loans ineligible for GSE securitization.

7.3.2 Aggregate home purchase effects by region

Another natural question is whether our predicted loan volume effects, incorporating all the heterogeneity we estimated as well as geographical variation in CS pre-policy, varies by zip code. If they do, we would be able to examine the house price responses to interest rates, and supply constraints at the zip code level would lead to a countervailing effect on home buyer demand. Unfortunately, as Figure A.6 shows, there is limited variation in predicted loan volume changes by zip code, with estimates bunching sharply at zero and having a very low standard deviation of around 1%, reflecting the relatively individualized nature of the policy. The limited scope for zip code level general equilibrium effects, which would have a countervailing effect on home buying, may be another reason for our larger estimated home buying elasticities.

7.3.3 Effects on loan size, home prices

We next examine the effect of the LLPA change led to changes in the intensive margin, such as prices of homes purchased and the loan size. Interestingly, we do not find statistically significant effects on the average price of homes purchased and the loan amount, shown in Table A.5 in Appendix F, consistent with the small intensive margin elasticity of loan demand to interest rates estimated in DeFusco and Paciorek (2017). There are increases in the average price of homes purchased and the loan amount among GSE loans, while the effects are negative and insignificant for non-agency conforming loans. The small overall effects may reflect some substitution between GSE loans and non-agency loans. This finding is also consistent with Bhutta and Ringo (2021), who find large home-purchase elasticities but small intensive margin elasticities. This finding is also consistent with a tight housing market, where home buyers have difficulties in getting matched to houses instead of optimizing over a set of many available houses with different price points.

7.3.4 Home purchase effects by lender type

Nonbanks account for a large share of originations sold to GSEs. We examine how the effects of the g-fee change differ by lender type in this section. Pass-through of the g-fee change to GSE loan costs is similar between banks and nonbanks (Table A.6 in Appendix G). However, for non-agency conforming loans, banks show no pass-through while nonbanks exhibit full pass-through, likely reflecting differences in funding sources: nonbanks sell to aggregators who may securitize through GSEs, while banks retain these loans on balance sheet.

This differential pass-through leads to an increase in the nonbank share of non-agency originations in response to the g-fee decrease (Table A.7 in Appendix G), while the GSE loan nonbank share remains stable. The similar responses in GSE loan originations across lender types provide further evidence against credit score manipulation, which would more likely occur at nonbanks facing less regulatory scrutiny (Buchak et al., 2018).

8 Model

We develop a structural model of frictional housing search for two purposes. First, we use the model to translate the reduced-form relative home purchase elasticities into the aggregate elasticity, taking into account the spillovers across homebuyers competing in the same housing market. Second, we evaluate how alternative g-fee designs might influence housing market outcomes with the model.

Our model introduces two key innovations. First, we endogenize both buyer entry and seller listing behavior by incorporating logits for buyer and seller flows, allowing us to capture how financing costs influence participation on both sides of the market. Second, we allow for buyers to enter housing search with heterogeneous reference interest rates determined by their mortgage choices, which then affect their competitiveness during the search. Our model has an excellent fit in sample, and incorporates many realistic features of the housing market with reasonable parameters.

8.1 Demand side

Our model has two stages. In the first stage, buyers and sellers decide to enter the housing market based on the expected future value of participating in the market. Buyers also choose their loan type in this stage, if they decide to enter, which in turn determines their reference interest rate in their housing search. In the second stage, buyers and sellers play a continuous-time search and matching game that is standard in the literature. This two-stage setup allows for endogenous flows of buyers and sellers into housing search models.

8.1.1 Entry

In the first stage, buyers, who differ in their credit score c , income bin k , choose a mortgage from lender j , at LTV l , and FHA vs GSE vs non-agency conforming mortgage f . Buy i 's choice maximizes:

$$u_{ijlf} = \gamma_{ck}^B V^B(r_{ckjlf}) + \xi_{cklfb(j)} + \epsilon_{ijlf}, \quad (3)$$

where $V^B(r_{ckjlf})$ is the expected value of searching for a home with reference interest rate r_{ckjlf} , $\xi_{cklfb(j)}$ captures systematic preferences for loan-type and lender-type (whether the lender is a bank or a non-bank $b(j)$) combinations, and ϵ_{ijlf} is an idiosyncratic taste shock. The choice of mortgage product (j, l, f) pins down the buyer's reference interest rate, which determines their competitiveness during the subsequent housing search stage.

The preference parameter related to the value of buying at a given interest rate γ_{ck}^B varies by credit score c and income group k as:

$$\gamma_{ck}^B = \gamma_0^B \exp(\gamma_c^b(c - \bar{c}) + \gamma_k^b(k - \bar{k}) + \gamma_{ck}^b(c - \bar{c})k), \quad (4)$$

where c is the credit score bin of the borrower divided by 100, with $\bar{c} = 7.2$, k is the income bin of the borrower from 0 (LMI) to 4 (90% percentile) and $\bar{k} = 2$. This specification allows the interest elasticity of home buying to vary flexibly by credit score, income, and their interaction.

Buyers use the outcome of the maximization in Equation (3), r_i , as their reference interest rate when deciding how much they value a house in the subsequent search and matching game. Households may also choose not to proceed with searching for a home at all, in which case they take the outside option with a utility equal to 0. The inflow of buyers of type r_i is then:

$$\mu^B(r_i) = \sum_{ckjlf} w_{ck} \mathbb{1}(r_i = r_{ckjlf}) \frac{\exp(\gamma_k^B V^B(r_{ckjlf}) + \xi_{cklfb(j)})}{1 + \sum_{jlf} \exp(\gamma_k^B V^B(r_{ckjlf}) + \xi_{cklfb(j)})} \quad (5)$$

where w_{ck} is the share of buyers with credit score c and income k in the population.

The flow of sellers similarly depends on the elasticity to the expected seller value V^S based on parameters γ_0^S, γ_1^S :

$$\mu^S = \frac{\exp(\gamma_0^S + \gamma_1^S V^S)}{1 + \exp(\gamma_0^S + \gamma_1^S V^S)}, \quad (6)$$

where γ_0^S pins down the mean utility of entering the market as a seller and γ_1^S pins down its sensitivity to the expected seller value V^S . In particular, γ_1^S is related to the general equilibrium responses to financing costs: if more buyers flow into the market in response to financing cost changes, γ_1^S pins down the level of seller response. If $\gamma_1^S = 0$, the number of transactions is

fixed at the initial number of sellers. On the other hand, if γ_1^S is large, the number of sellers becomes perfectly elastic, and the number of transactions is entirely pinned down by buyer inflows.

8.1.2 Housing search

In the second stage, buyers and sellers play a continuous-time matching game. Buyers value the house at ρ^b , but their value may be reduced by their reference interest rate by the amount of ψ per 100 bps of interest rate. This reduction in value may reflect both the disutility of higher mortgage payments as well as any DTI supply constraints. Upon matching and paying price P , which we endogenize, the buyer's surplus is:

$$V^B(r_i) = \rho^b - \psi r_i - P + \epsilon_b, \quad (7)$$

where buyers draw $\epsilon_b \sim N(0, \sigma)$ each time they match, representing an idiosyncratic match quality.

For sellers, the surplus for a given price P and transaction costs τ_s is:

$$V^S = P - \tau_s P. \quad (8)$$

A buyer with a higher r_i has less surplus and therefore their offer is less attractive to the seller, but if the buyer has a high match quality (ϵ_b), they may still make an attractive offer despite high r_i . Crucially for our purposes, σ corresponds to price dispersion, whereas ψ relates to the semi-elasticity of relative home purchases to g-fee changes in a competition channel. The reason why the effect of ψ is primarily driven by a competition channel is that, holding fixed seller flows, the total number of transactions is pinned down by the number of sellers, and so a larger ψ makes it more likely for a home buyer with a lower interest rate to win the home and crowd out other home buyers.

Endogenous masses of buyers M^B and sellers M^S are available for matching. The matching

function is:

$$m(M^B, M^S) = \alpha \left(\sum_i M^B(r_i) \right)^\phi (M^S)^{1-\phi} \quad (9)$$

where α represents the overall matching rate, and ϕ represents how imbalances in M^B, M^S affects the match rate.

The equilibrium price is determined by Nash bargaining with the seller's bargaining strength being θ , where the surplus is split. The price conditional on trade is:

$$P(\epsilon_b, r_i) = \frac{1}{1 + \theta\tau_s} \left(V^S + \theta(\rho^b - \psi r_i + \epsilon_b - V^B(r_i) - V^S) \right) \quad (10)$$

Both buyers and sellers are forward-looking. Buyers discount the future at rate δ and exit at rate κ , capturing exit in response to inability to match with a home. Sellers are assumed to keep their homes on the market forever, but discount the future at a rate $\delta + \delta^s$, with a higher discount rate capturing sellers' urgency to sell. The seller's urgency to sell δ^s helps pins down housing search moments such as days on market.

8.2 Supply side

8.2.1 Lenders

Lender j maximizes profits by setting rates separately for each credit score bin c , income bin k , LTV l , and loan type f :

$$\max_{\mathbf{r}_j} \sum_{cklf} (Mr_{jcklf} - c_{jcklf}) s_{jcklf} \quad (11)$$

where $\mathbf{r}_j$ is the vector collecting mortgage rates, M is a conversion of factor of rates from fees, and s_{jcklf} represents the lender's market share for that specific loan product. The cost c_{jcklf} for GSE loans, which we label as loans with $f = 0$, is:

$$c_{jckl0} = LLPA_{cl} + \tilde{c}_{jckl0} \quad (12)$$

which captures lenders' costs upon GSE delivery, with $\tilde{c}_{jckl0}$ representing any residual non-LLPA delivery costs such as origination or servicing costs.

Writing $\mathbf{c}_j$ and $\mathbf{s}_j$ as vectors collecting costs and market shares across c, k, l, f , and Δ being the Jacobian of s_{jcklf} with respect to $\mathbf{r}_j$, $\mathbf{s}_j$ being a vector representing the s_{jcklf} , the first order condition is:

$$M\mathbf{r}_j - \mathbf{c}_j = -M\Delta_j^{-1}\mathbf{s}_j \quad (13)$$

We assume that there are N^{nb} non-bank lenders and N^b bank lenders operating in each market, with lenders being identical within each type in terms of their marginal costs $\mathbf{c}_j$.

8.2.2 GSEs

In the model estimation, we assume that the GSEs set LLPAs exogenously, as $LLPA_{lc0}$, and thus the GSEs do not have an objective function. In the counterfactual, we calculate the GSE profits as follows:

$$\pi^{GSE} = \sum_{lc} s_{lc}^{GSE} (LLPA_{lc} - ELOSS_{lc}) \quad (14)$$

where $s_{lc}^{GSE} = \sum_{jk} s_{lcjk}^{GSE} w_{ck}$, and $ELOSS_{lc}$ represents the expected losses for those loans.

8.3 Model solution

The model solution is given in Proposition 1.

Proposition 1. *A stationary equilibrium of this model is described by the seller and buyer value functions $V^S, V^B(r_i)$, the seller and buyer masses $M^S, M^B(r_i)$, matching rates λ^S, λ^B , and trade cutoff function $\bar{\epsilon}(r_i)$ with $G(r_i) = \Pr(\epsilon_s > \bar{\epsilon}(r_i))$, which satisfies the following conditions:*

Trade cut-offs (condition for which a transaction occurs following a match):

$$\epsilon_b \geq V^S + V^B(r_i) - (\rho^b - \psi r_i) = \bar{\epsilon}(r_i) \quad (15)$$

Matching rates:

$$M^S \lambda^S = \sum_i M^B(r_i) \lambda^B = \alpha \left(\sum_i M^B(r_i) \right)^\phi M_S^{1-\phi} \quad (16)$$

Flow equality:

$$M^B(r_i) \lambda^B G(r_i) + \kappa M^B(r_i) = \mu^B(r_i) \quad (17)$$

$$M^S \lambda^S \frac{\sum_i M^B(r_i) G(r_i)}{\sum_i M^B(r_i)} = \mu^S \quad (18)$$

where $\mu^B(r_i)$ satisfies Equation (5) and μ^S satisfies Equation (6).

Bellman equations:

$$(\delta + \kappa) V^B(r_i) = \lambda^B \left(1 - \frac{\theta}{1 + \theta \tau_s} \right) \int_{\epsilon_b > \bar{\epsilon}(r_i)} (\rho^b - \psi r_i + \epsilon_b - V^B(r_i) - V^S) dN(\epsilon_b) \quad (19)$$

$$(\delta + \kappa^S) V^S = \lambda^S \frac{\theta}{1 + \theta \tau_s} \sum_i \frac{M^B(r_i)}{\sum_i M^B(r_i)} \int_{\epsilon_b > \bar{\epsilon}(r_i)} (\rho^b - \psi r_i + \epsilon_b - V^B(r_i) - V^S) dN(\epsilon_b) \quad (20)$$

Proof. The trade-cutoffs in Equation (15) come from the match surplus term being positive in Equation (10). The rest follows the definition of a stationary equilibrium. $\square$

This model can be solved numerically, with two outer loops. The computational details of the model are shown in Appendix Section H. The expected price is:

$$\bar{P} = \sum_i \frac{M^B(r_i)}{\sum_i M^B(r_i)} \int P(\epsilon_b, r_i) M^B(r_i) \lambda^B G(r_i) \mathbb{1}(\epsilon_b > \bar{\epsilon}(r_i)) dN(\epsilon_b) \quad (21)$$

8.4 Identification

8.4.1 Externally calibrated parameters

We obtain $\delta = 0.05, \alpha = 1$ from Gerardi (2017). We then obtain $ELOSS_{lc}$ from the credit risk transfer (CRT) model-implied LLPA grid from Capponi et al. (2026), which we take as

the private-capital equivalent baseline for pricing GSE credit risk.¹⁴ Finally, we obtain the rate-to-fee conversion factor $M = 1/0.406$, with 0.406 being from our calibration of the MBS TBA coupon increase needed for 1% of loan balance in value over 2022–2023, calibrated to the current coupon of the Primary Mortgage Market Survey (PMMS) rate minus a 25bps servicing fee and a 43bps uniform on-going g-fee, following the approach in [Fuster et al. \(2025\)](#).

8.4.2 Internally calibrated parameters

For the internally calibrated parameters, while all parameters affect the model jointly, intuitively each of them captures different variation in the data. Specifically, the calibration targets include:

- ξ_{cklf} : to market shares of that lender/product type by income and credit score,
- ρ^b : to the average house price, equal to \$432,950 for 2023,
- γ_0^B : to the elasticity of substitution between bank and non-bank lenders following g-fee changes, equal to 0.019 from Appendix Table A.7,
- ψ : to the total home purchase elasticity to g-fee changes, equal to 13.9% from Table 3.
- σ : to the idiosyncratic log standard deviation of house prices, equal to 9.3% from [Gerardi et al. \(2024\)](#),
- κ^s : to the sellers' days on market (DOM), equal to its 2023 value of 52, from Realtor.com,
- κ : to the number of buyer offers before success, equal to 2.5, from NAR survey of home buyers,¹⁵
- ϕ : to the number of weeks buyers search for their homes, equal to 12, from NAR survey of home buyers.¹⁶

We now discuss the calibration of the remaining parameters of γ_0^S, γ_1^S . The sellers' flow rate responses to the value of being a seller, γ_1^S , is not identified in our data. We use a fixed point

¹⁴[Capponi et al. \(2026\)](#) finds an approximately 50bps of GSE markups on top of their model-implied LLPA grid per dollar of loan amount, which implies that the GSEs made about \$3 billion in profit on their \$616 billion new purchase originations in 2023. The fact that this profit is feasible likely stems from the value of the implicit government guarantee in extreme rare-disaster scenarios as well as increasingly stringent bank capital regulation from BASEL III which limits non-agency options.

¹⁵Available [HERE](#).

¹⁶Available [HERE](#).

to calibrate the seller flow rate μ^S to a buyer success probability of 40%.¹⁷ Since we calibrated the buyer outside option to 90%, our assumptions imply that 10% of households are searching for a home to buy and 4% of households ends up being successful in each time period, which is realistic based on Gerardi et al. (2024). This $\mu^S = 0.04$ in turn implies a value of γ_0^S for each assumed level of γ_1^S , which allows us to solve our model counterfactuals under an assumption about γ_1^S . For our baseline analyses, we assume that $\gamma_1^S = 0$, which implies that the number of transactions is fixed at the initial number of sellers. We also examine the sensitivity of our results to allowing fully flexible seller flows, implicitly assuming that $\gamma_1^S \rightarrow \infty$ and that the number of home purchases is purely pinned down by buyer demand. We view the $\gamma_1^S = 0$ and $\gamma_1^S \rightarrow \infty$ as two extremes captures the range of variation across U.S. geographies.

8.5 Calibration

Table 6 shows the internally calibrated model parameters.

Table 6: Calibrated model parameters

Parameter	Description	Value
κ	Buyer exit rate	0.2853
δ^s	Seller discount rate increment	1.1585
ψ	Mortgage rate weight in flow payoff	13.3757
ρ^b	Buyer flow payoff from owning	65.1360
σ	Match shock std. dev.	14.1292
ϕ	Matching function elasticity (buyers)	0.7709
γ_0^B	Rate sensitivity (multiplier)	0.0374
γ_c^b	Rate sensitivity (credit score gradient)	3.7785
γ_k^b	Rate sensitivity (income gradient)	0.6341
γ_{ck}^b	Rate sensitivity (credit score by income gradient)	-1.1029

Among our calibration parameters, κ implies that the buyers drop out at a rate of 0.28 per year. A $\delta^s = 1.1585$ implies a seller discount rate of $1.1585 + 0.05 = 1.2085$ per year, or approximately 6.8% monthly, which reflects the sellers' high average urgency to sell and helps fit DOM moments given the observed idiosyncratic price dispersion of house prices. Our ψ

¹⁷Based on an industry pre-approval to application conversion rate from LenderLogix of 58% [HERE](#) multiplied by a application to origination rate of 69%.

implies that that a 1 percentage point rise in interest rate reduces the buyers' valuation of the house by $13.3757/65.1360=20.5\%$, which may be reasonable because an interest rate increase from 5% to 6% raises the monthly mortgage payment by 11.6%, plus the buyer may become DTI constrained. An alternative interpretation of ψ is that, for a 100 bps change in interest rate ($\Delta r = 1.00$), the seller becomes $\psi/\sigma = 13.4572/13.7576 = 0.95$ standard deviations more willing to accept the buyer's offer, which could capture the different ways the buyer made their offer better. Our σ estimate reflects the idiosyncratic variation in sellers' urgency to sell and is used to fit price dispersion moments. Our $\phi = 0.7709$ is calibrated to housing search moments and is consistent with the 0.84 value typically estimated in the literature such as [Genesove and Han \(2012\)](#). Finally, $\gamma_0^B, \gamma_c^b, \gamma_k^b, \gamma_{ck}^b$ are calibrated to non-bank substitution rates as well as the relative home purchase semi-elasticities we estimated in Section 7.

Table 7 shows the model's fit to targeted housing market search and lender type choice moments. The model fits the targeted moments well, in terms of the buyer's expected search time before success, sellers' DOM, the buyers' expected number of offers before success, the expected price, the idiosyncratic price dispersion in terms of the standard deviation of log price, and our estimated relative home purchase semi-elasticities as well non-bank share changes to a 100 bps g-fee rise.

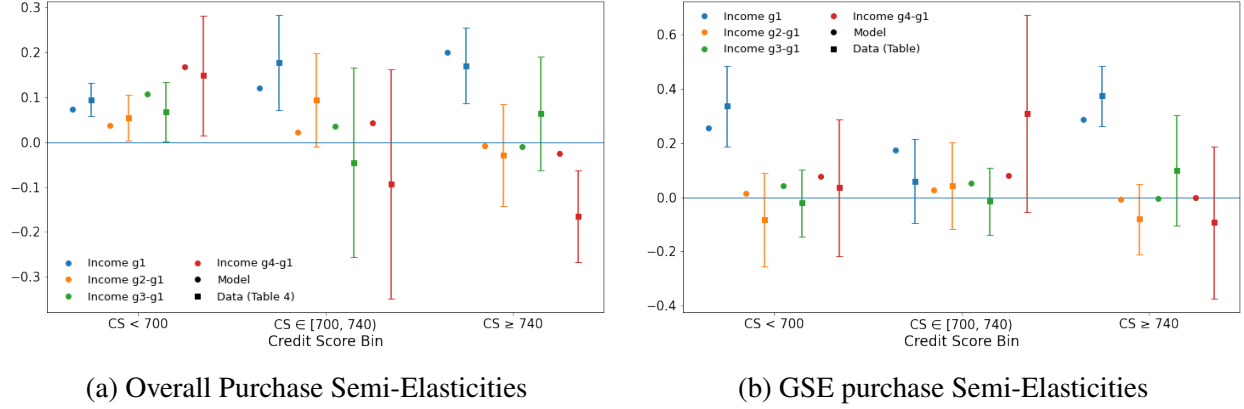
Table 7: Comparison of model output to targeted moments

Statistic	Model Output	Target	Difference
Buyer weeks to success	9.924	12.000	-2.076
Seller days on market (DOM)	51.749	52.000	-0.251
Expected number of offers before success	2.675	2.500	0.175
Average house price (\$10,000s)	43.534	43.295	0.239
Std. dev. of log price	0.096	0.093	0.003
Relative home purchase semi-elasticity (100 bps g-fee)	0.147	0.139	0.008
Non-bank share change (100 bps g-fee)	0.013	0.019	-0.006

Figure 5 examines the fit of our model to overall and GSE purchase semi-elasticities by credit score group and income bin, from Tables 3 and analogous estimates for GSE loans, which we also targeted. Despite having only three parameters governing preference heterogeneity, consisting of a linear effect on credit score bin, income bin, and an interaction term, the model

fits the full set of heterogeneous semi-elasticities we estimate well.

Figure 5: Fit of our model to overall and GSE purchase semi-elasticities by income and CS



Note: Round dots represent model predictions, whereas square dots with error bars represent estimates from Table A.4 in Figure 5(a) and analogous estimates for GSE loans only in Figure 5(b).

Panel A of Table 8 examines the model's implications for several untargetted moments. The first row of panel A of Table 8 shows that the intensive margin elasticity of the home price that a borrower experiencing a +100 bps idiosyncratic change in interest rate purchases is only -3%, consistent with a similarly small -2% estimate from the conforming-jumbo spread in DeFusco and Paciorek (2017). This implies that the main effect of higher interest rates is in making households less able to buy a home, rather than reducing their transaction price conditional on a successful purchase. The second row of panel A of Table 8 shows that the expected relative change in CS<680 share following a 50 bps FHA MIP cut in our model is 12.4%, which is similar to the 14% estimate of Bhutta and Ringo (2021). The third row of panel A of Table 8 shows the aggregate growth in home sales following a uniform 50 bps increase in rate. The effect of this counterfactual depend strongly on our assumption about seller supply, unlike our previous May 2023 LLPA change counterfactual and the 50 bps FHA MIP cut counterfactual for which seller supply assumptions have minimal effects on the moments reported.¹⁸ We

¹⁸Specifically, our model-implied May 2023 g-fee elasticity of 14.7% is unchanged from the baseline fixed seller supply estimate of 14.7% when we switch to allowing free seller entry, with $\gamma_1^S \rightarrow \infty$, and our non-bank share change (100 bps g-fee) barely increases from 0.012 to 0.013 when we switch to allowing free seller entry. These small effects suggest that our model predicts limited effects of the May 2023 LLPA change on aggregate demand regardless of seller entry, which is also discussed in Section 9. Similarly, our model-implied relative growth in CS<680 volume following

therefore report both the baseline model estimate of 0, with fixed seller supply, and 0.164, with free seller entry and $\gamma_1^S \rightarrow \infty$, for this counterfactual, in the Model Output column. With the two supply scenarios as lower and upper bounds, we obtain a range of 0–0.164 as shown in the third row of Table 8, which brackets the aggregate estimate of 0.134 from Table 6, column 1 of [Hacamo \(Forthcoming\)](#).

Our model-implied relative home purchase elasticities to a 100 bps change in g-fee represent a mix of two sets of parameters: $\gamma_0^B, \gamma_c^b, \gamma_k^b, \gamma_{ck}^b$, which determines the inflow rate of buyers as they respond to their expected value of housing search conditional on the interest rate, and ψ during the housing search. Panel B of Table 8 examines the relative importance of these two channels by shutting down the former and keeping buyer flows in Equation (5) constant, while applying the success probabilities of the May 2023 LLPA change in partial equilibrium. The result is a 9.4% relative home purchase elasticity at our CS cut-offs, as opposed to 14.7% in the baseline model, suggesting about two thirds of the relative home purchase elasticity is due to a competition effect rather than an entry effect. This feature of the model is also consistent with our ability to fit the lower aggregate interest rate elasticity estimated in [Hacamo \(Forthcoming\)](#) as an untargeted moment.

Table 8: Model implications for untargeted moments

Panel A: Untargeted moments from the literature

Statistic	Model Output	Data
Intensive margin elasticity (+100 bps interest rate)	-0.030	-0.020
Relative CS<680 growth following a 50 bps FHA MIP cut	0.124	0.140
Aggregate growth in home sales following a 100 bps increase in rates	0–0.164	0.134

Panel B: Implied estimates with constant flows (only competition effect)

Statistic	Model Output
Relative home purchase semi-elasticity (100 bps g-fee)	0.094

a 50 bps FHA MIP cut with free seller entry barely changes at 0.124, compared to 0.123 in the baseline case, consistent with the specialized nature of the FHA market that occupies a lower-competition high interest rate region in the model.

9 Counterfactuals

We use our calibrated model to evaluate three counterfactual exercises. First, we examine whether GSEs can increase their capitalization by raising g-fees. Second, we simulate the effects of increasing g-fees in a looser housing market with higher days on market. Third, we simulate the effects of the May 2023 LLPA reform to quantify its distributional and profit consequences. For each exercise, we report results under both fixed seller supply ($\gamma_1^S = 0$) and free entry ($\gamma_1^S \rightarrow \infty$), which we view as bracketing the range of supply-side responses across U.S. housing markets. We also report estimates with a γ_0^S, γ_1^S calibrated to [Hacamo \(Forthcoming\)](#)'s aggregate interest rate elasticity in Appendix Section K, which in general leads to outcomes in between our main estimates.

9.1 Uniform increases to g-fees

Our first counterfactual examines the implications of a uniform increase in upfront g-fees of 25 basis points across all FICO-LTV cells. Such an increase in g-fees is often contemplated to improve GSE capitalization and increase Treasury income.¹⁹ As the GSEs prepare to exit conservatorship, they may internalize more of a profit motive, and a uniform g-fee increase represents the simplest policy instrument available.

We find that under both fixed supply and free entry scenarios, increases in g-fees causes some groups to decrease their home purchases and others to increase, with the latter being due to the competition channel. Table 9 reports the results. Under fixed supply in Panel A, the 25 bps increase raises average markups from 51 to 76 bps. Total home purchases are unchanged under fixed supply by construction, but there is a mild reshuffling: low-FICO purchases rise modestly (e.g., +2.73% for FICO 620) while high-FICO purchases decline slightly (−1.39% for FICO 780), as the uniform g-fee increase narrows the relative pricing advantage of GSE loans for high-FICO borrowers relative to FHA mortgages. GSE volume declines more for high-FICO borrowers, reflecting the fact that high-FICO borrowers substitute more to other products. GSE profits rise from \$3.17 billion to \$4.56 billion, an increase of \$1.39 billion,

¹⁹See a report by the CBO [HERE](#).

indicating that the volume decline is more than offset by the higher markups.

Under free entry in Panel B, the effects on total home purchases are amplified, as the competition channel weakens. Total home purchases fall by 2.63%, with high-FICO groups experiencing the largest declines (−5.61% for FICO 780) and low-FICO groups experiencing near-zero or small positive changes (+1.08% for FICO 620). GSE volume declines by 5.9% in aggregate, roughly double the fixed-supply decline, as the reduction in buyer demand feeds back into fewer seller listings. Due to the larger extensive margin effects, GSE profits rise by \$1.26 billion, slightly less than under fixed supply due to the larger volume contraction.

Table 9: Effect of Uniform G-Fee Increase (+25 bps) on GSE Markups and Profits

FICO	Δ Purch.%	Δ GSE%	Markup (bps)			Profit (\$B)		$\Delta\pi$ (\$B)
			Old	New	Δ	Old	New	
<i>Panel A: Fixed Seller Supply</i>								
780	−1.39	−5.1	56	81	+25	1.14	1.57	+0.43
760	−0.54	−3.1	51	76	+25	0.56	0.81	+0.25
740	−0.16	−2.1	51	76	+25	0.47	0.69	+0.22
720	+0.04	−1.8	45	70	+25	0.33	0.50	+0.17
700	+0.40	−1.5	41	66	+25	0.23	0.37	+0.13
680	+0.87	−1.5	51	76	+25	0.20	0.29	+0.09
660	+1.54	−1.4	51	76	+25	0.11	0.16	+0.05
640	+1.98	−1.4	53	78	+25	0.06	0.09	+0.03
620	+2.73	−1.6	83	108	+25	0.06	0.07	+0.02
Total	−0.00	−3.1	51	76	+25	3.17	4.56	+1.39
<i>Panel B: Free Entry</i>								
780	−5.61	−9.1	56	81	+25	1.14	1.51	+0.36
760	−3.45	−5.9	51	76	+25	0.56	0.79	+0.23
740	−2.37	−4.3	51	76	+25	0.47	0.67	+0.20
720	−1.85	−3.7	45	70	+25	0.33	0.49	+0.16
700	−1.35	−3.3	41	66	+25	0.23	0.36	+0.13
680	−0.80	−3.3	51	76	+25	0.20	0.29	+0.09
660	+0.00	−3.2	51	76	+25	0.11	0.16	+0.05
640	+0.48	−3.2	53	78	+25	0.07	0.09	+0.03
620	+1.08	−3.6	83	108	+25	0.06	0.07	+0.01
Total	−2.63	−5.9	51	76	+25	3.17	4.43	+1.26

Notes: This table reports outcomes under a uniform 25 basis point increase in upfront g-fees across all FICO-LTV cells. Markups are LLPA minus model-implied cost, in basis points. Profits are markup $\times$ GSE origination volume, scaled to \$616B total GSE originations. Old markups and profits correspond to the post-May 2023 LLPA schedule. Panel A holds seller supply fixed; Panel B allows seller entry proportional to buyers.

Appendix Table [A.8](#) presents the income-group implications. Under fixed supply, the uniform g-fee increase has negligible effects on home purchases across all income groups, with changes ranging from +0.11% for LMI borrowers to −0.35% for high-income borrowers. Under free entry, all income groups experience declines of 2.4–3.2%, with higher-income groups declining slightly more due to their greater reliance on GSE loans. Overall, these results suggest that a modest uniform g-fee increase can generate meaningful additional earnings for the GSEs, and that even though our estimated elasticities are large, they are not large enough to make marginal increases to g-fees unprofitable for the GSEs.

9.2 Uniform increases to g-fees in less tight housing markets

Our second counterfactual examines further increases to g-fees when markets have a higher DOM. To do so, we increase the baseline flow of seller entry until we reach a DOM of 64 days, equivalent to the 2019 pre-pandemic average.²⁰ The results are shown in Table [10](#). Generally, the results are consistent with a weakened competition effect, which is sensible due to a larger baseline supply of sellers. Column 1 of Panel A of Table [10](#), which is the fixed supply case, shows smaller magnitudes of home purchase effects in both directions compared to Panel A of Table [9](#). The reduction in magnitudes reflect a weakened competition effect under fixed seller supply.

Column 1 of Panel B of Table [10](#), which refers to the free entry case, also shows a weaker competition effect compared to Panel B of Table [9](#). For lower credit score borrowers, the increase in purchases of +0.39% at FICO 620 is lower than the +1.08% in the tighter market case. For high credit score borrowers, the decline in home purchases is larger, suggesting a wider range of potential outcomes for this group of borrowers depending on seller supply response assumptions. The results on GSE profits are between \$1.24–\$1.40 billion, similar to those in the baseline case shown in Table [9](#), suggesting that the GSEs can still increase their capitalization in higher DOM markets by increasing g-fees.

Appendix Table [A.9](#) presents the income-group implications. Under fixed supply, the effects on home purchases remain negligible across all income groups, ranging from +0.06% for LMI

²⁰The DOM data is obtained from the Realtor.com Housing Inventory Core Metrics [HERE](#).

borrowers to -0.19% for high-income borrowers, roughly half the magnitudes of the baseline market. Under free entry, all income groups experience declines of $3.0\text{--}3.9\%$, somewhat larger than the $2.4\text{--}3.2\%$ declines in the baseline market, with higher-income groups continuing to decline more due to their greater reliance on GSE loans.

Table 10: Effect of Uniform G-Fee Increase (+25 bps) in a Higher-DOM Market (DOM = 64)

FICO	Δ Purch.%	Δ GSE%	Markup (bps)			Profit (\$B)		$\Delta\pi$ (\$B)
			Old	New	Δ	Old	New	
Panel A: Fixed Seller Supply								
780	-0.75	-5.1	56	81	+25	1.12	1.54	+0.42
760	-0.24	-2.9	51	76	+25	0.55	0.80	+0.25
740	-0.14	-1.9	51	76	+25	0.47	0.69	+0.22
720	-0.06	-1.5	46	71	+25	0.34	0.52	+0.18
700	+0.16	-1.2	41	66	+25	0.24	0.38	+0.14
680	+0.43	-1.1	51	76	+25	0.21	0.31	+0.10
660	+0.84	-1.0	50	75	+25	0.11	0.16	+0.05
640	+1.07	-1.0	53	78	+25	0.06	0.09	+0.03
620	+1.45	-1.1	84	109	+25	0.06	0.08	+0.02
Total	-0.00	-2.9	51	76	+25	3.17	4.57	+1.40
Panel B: Free Entry								
780	-7.16	-11.2	56	81	+25	1.12	1.45	+0.32
760	-4.16	-6.7	51	76	+25	0.55	0.77	+0.22
740	-2.62	-4.4	51	76	+25	0.47	0.67	+0.20
720	-1.83	-3.2	46	71	+25	0.34	0.51	+0.17
700	-1.27	-2.6	41	66	+25	0.24	0.38	+0.14
680	-0.80	-2.4	51	76	+25	0.21	0.31	+0.10
660	-0.22	-2.2	50	75	+25	0.11	0.16	+0.05
640	+0.08	-2.2	53	78	+25	0.06	0.09	+0.03
620	+0.39	-2.4	84	109	+25	0.06	0.07	+0.02
Total	-3.19	-6.4	51	76	+25	3.18	4.41	+1.24

Notes: This table reports outcomes under a uniform 25 basis point increase in upfront g-fees across all FICO-LTV cells, in a market recalibrated to an average days-on-market of 64 days (the 2019 pre-pandemic average) by raising the baseline flow of seller entry. Markups are LLPA minus model-implied cost, in basis points. Profits are markup $\times$ GSE origination volume, scaled to \$616B total GSE originations. Old markups and profits correspond to the post-May 2023 LLPA schedule. Panel A holds seller supply fixed; Panel B allows seller entry proportional to buyers.

9.3 Simulating the effects of the May 2023 g-fees change

Our third counterfactual simulates the effects of the May 2023 LLPA reform. Table 11 reports the results. In the second column of Panel A of Table 11, under fixed seller supply, the reform increases home purchases for the highest-FICO borrowers (780) by 3.5% and for low-FICO borrowers (620–660) by 1.2–4.1%, reflecting the LLPA reductions these groups received. Mid-FICO borrowers (700–760) reduce their home purchases by between 0.6% and 4.3%, with the largest losses concentrated at 720–740, where the markup increases were steepest. The total number of purchases is unchanged by construction when $\gamma_1^S = 0$. The third column of Panel A of Table 11 reveals that effects on GSE origination volume are larger than the effects on total purchases, as borrowers also substitute across loan types. GSE volume rises by 12% to 19% for low-FICO borrowers (620–660) and by 5.2% for FICO 780, while falling by 1.0% to 6.2% for mid-FICO borrowers (680–760). In aggregate, total GSE volume rises by a modest 0.1%.

The fourth to fifth columns of Panel A of Table 11 suggest that the new LLPA schedule compresses the range of GSE markups over $ELOSS_{lc}$. Under the old LLPA schedule, GSE markups ranged from 15 bps at FICO 720 to 135 bps at FICO 620; the new schedule has a range of 46 bps to 86 bps, moving closer to risk-based pricing with a constant markup, which is as found in Capponi et al. (2026). Finally, the sixth to eighth columns of Panel A of Table 11 show that aggregating across FICO bins, the reform raised average GSE markups by 4 bps and also increased GSE profits modestly, by \$0.24 billion, from \$2.92 billion to \$3.17 billion.

Under free entry in Panel B of Table 11, the patterns across FICO groups are broadly similar. The difference is that, as the second column in Panel B of Table 11 shows, total purchases increase by 0.21% rather than remaining fixed, and the third column in Panel B of Table 11 shows that total GSE volume rises by 0.3% rather than 0.1%. The profit implications are nearly identical, with the GSEs gaining \$0.25 billion.

Appendix Table A.10 translates these effects into income-group impacts. Under both fixed supply and free entry, the reform has negligible effects across all income groups, consistent with the fact that income is only weakly correlated with credit scores (Beer et al., 2018). Overall, our results suggest that the May 2023 change only had modest effects on overall home purchases, had some effects on borrower demand in line with increased alignment of prices with credit

Table 11: Effect of May 2023 G-Fee Change on GSE Markups and Profits

FICO	Δ Purch.%	Δ GSE%	GSE Markup (bps)			GSE Profit (\$B)		$\Delta\pi$ (\$B)
			Old	New	Δ	Old	New	
<i>Panel A: Fixed Seller Supply</i>								
780	+3.45	+5.2	66	56	−10	1.29	1.14	−0.14
760	−1.93	−2.8	41	51	+10	0.47	0.56	+0.09
740	−4.29	−6.2	24	52	+28	0.23	0.47	+0.24
720	−4.06	−6.1	15	46	+31	0.12	0.33	+0.21
700	−0.60	−1.0	38	42	+4	0.21	0.23	+0.02
680	−0.57	−1.1	46	51	+6	0.18	0.20	+0.02
660	+4.11	+14.9	111	50	−61	0.21	0.11	−0.10
640	+3.31	+19.1	128	54	−73	0.13	0.06	−0.07
620	+1.22	+12.3	135	86	−49	0.08	0.06	−0.03
Total	+0.00	+0.1	47	52	+4	2.92	3.17	+0.24
<i>Panel B: Free Entry</i>								
780	+3.72	+5.4	66	56	−10	1.29	1.14	−0.14
760	−1.73	−2.6	41	51	+10	0.47	0.56	+0.09
740	−4.11	−6.1	24	52	+28	0.23	0.47	+0.24
720	−3.88	−6.0	15	46	+31	0.12	0.33	+0.21
700	−0.41	−0.8	38	42	+4	0.21	0.23	+0.02
680	−0.39	−1.0	46	51	+6	0.18	0.20	+0.02
660	+4.30	+15.1	111	50	−61	0.21	0.11	−0.10
640	+3.49	+19.4	128	54	−73	0.13	0.07	−0.07
620	+1.42	+12.5	135	86	−49	0.08	0.06	−0.02
Total	+0.21	+0.3	47	52	+4	2.92	3.17	+0.25

Notes: This table reports the effect of the 2023 LLPA reform. Δ Purch.% is the change in total home purchases across all channels. Δ GSE% is the change in GSE origination volume. GSE markups are actual LLPA minus model-implied cost, in basis points. Profits are markup $\times$ GSE origination volume, scaled to \$616B total GSE originations. Panel A holds seller supply fixed; Panel B allows sellers to enter proportionally with buyers.

risk, and improved GSE capitalization. This conclusion is in sharp contrast to the media controversy that accompanied the change.²¹

10 Conclusion

The elasticity of home purchases to financing costs is important but understudied. In this paper, we provide some of the first natural experiment evidence on it that applies to a broad population by leveraging a series of g-fee changes that vary sharply across credit score bins. We find large effects on relative home purchases by the affected populations, particularly in tight housing markets, and in a manner that is not fully explained by debt-to-income (DTI) constraints. These results support models of the housing market that incorporate the direct effects of financing costs on household demand as well as supply constraints in general equilibrium.

An important implication of our reduced form results is that policies that influence household financing costs in relatively individualized ways, including the GSEs' pricing policy, can have significant effects on individual home purchases. Other such policy includes interest rate subsidies for veterans as well as from certain private organizations such as universities. Our results imply that interest rate subsidy programs can effectively increase the home purchases of a broad set of households, particularly in tight housing markets.

Our structural model builds upon and extends our reduced form results. An important implication of our structural model is that as GSEs prepare for reform, they can increase g-fees to improve their capitalization. That is, even though our estimated home purchase elasticities and estimated substitution to other loan types in response to g-fee increases are large, they are not large enough to make further increases to g-fees unprofitable. This result is useful for ongoing policy discussions regarding GSE reform.

²¹See, e.g., [ABC News](#).

References

- Adelino, Manuel, and Taha Ahsin, 2025, Does local competition matter for mortgage pricing? evidence from fee pass-through, *Working paper* .
- Adelino, Manuel, Antoinette Schoar, and Felipe Severino, 2025, Credit supply and house prices: Evidence from mortgage market segmentation, *Journal of Financial Economics* 163, 103958.
- Albrecht, James, Axel Anderson, Eric Smith, and Susan Vroman, 2007, Opportunistic matching in the housing market, *International Economic Review* 48, 641–664.
- Amornsiripanitch, Natee, and Judith Ricks, 2024, Lenders’ response to mortgage guarantee fees, *Available at SSRN* .
- Anenberg, Elliot, and Patrick Bayer, 2020, Endogenous sources of volatility in housing markets: The joint buyer–seller problem, *International Economic Review* 61, 1195–1228.
- Beer, Rachael, Felicia Ionescu, and Geng Li, 2018, Are income and credit scores highly correlated?, *FEDS Notes* .
- Benson, David, You Suk Kim, and Karen M Pence, 2024, Nonbank issuers and mortgage credit supply, *Available at SSRN* 4615551 .
- Berry, Steven, James Levinsohn, and Ariel Pakes, 1995, Automobile prices in market equilibrium, *Econometrica* 63, 841–890.
- Bhutta, Neil, Andreas Fuster, and Aurel Hizmo, 2026, Paying too much? borrower sophistication and overpayment in the us mortgage market, *The Journal of Finance* 81, 49–90.
- Bhutta, Neil, and Daniel Ringo, 2021, The effect of interest rates on home buying: Evidence from a shock to mortgage insurance premiums, *Journal of Monetary Economics* 118, 195–211.
- Bosshardt, Joshua, Marco Di Maggio, Ali Kakhbod, and Amir Kermani, 2024a, The credit supply channel of monetary policy tightening and its distributional impacts, *Journal of Financial Economics* 160, 103914.
- Bosshardt, Joshua, Ali Kakhbod, and Amir Kermani, 2024b, The value of intermediaries for gse loans, Technical report, National Bureau of Economic Research.
- Buchak, Greg, Gregor Matvos, Tomasz Piskorski, and Amit Seru, 2018, Fintech, regulatory arbitrage, and the rise of shadow banks, *Journal of financial economics* 130, 453–483.
- Capponi, Agostino, Stijn Van Nieuwerburgh, and Xinkai Wu, 2026, Pricing residential mortgage credit risk in the post-gfc era, Technical report, National Bureau of Economic Research.
- Cengiz, Doruk, Arindrajit Dube, Attila Lindner, and Ben Zipperer, 2019, The effect of minimum wages on low-wage jobs, *The Quarterly Journal of Economics* 134, 1405–1454.
- Cohn, Jonathan B, Zack Liu, and Malcolm I Wardlaw, 2022, Count (and count-like) data in finance, *Journal of financial economics* 146, 529–551.

- Collier, Benjamin, and Cameron Ellis, 2024, A demand curve for disaster recovery loans, *Econometrica* 92, 713–748.
- Collier, Benjamin L, Daniel A Hartley, Benjamin J Keys, and Jing Xian Ng, 2024, Credit when you need it, Technical report, National Bureau of Economic Research.
- DeFusco, Anthony A, and Andrew Paciorek, 2017, The interest rate elasticity of mortgage demand: Evidence from bunching at the conforming loan limit, *American Economic Journal: Economic Policy* 9, 210–240.
- Elenev, Vadim, Tim Landvoigt, and Stijn Van Nieuwerburgh, 2016, Phasing out the gses, *Journal of Monetary Economics* 81, 111–132.
- Fuster, Andreas, Stephanie H Lo, and Paul S Willen, 2025, The time-varying price of financial intermediation in the mortgage market, *Journal of Finance* .
- Gabrovski, Miroslav, and Victor Ortego-Marti, 2019, The cyclical behavior of the beveridge curve in the housing market, *Journal of Economic Theory* 181, 361–381.
- Gabrovski, Miroslav, and Victor Ortego-Marti, 2021, Search and credit frictions in the housing market, *European Economic Review* 134, 103699.
- Genesove, David, and Lu Han, 2012, Search and matching in the housing market, *Journal of urban economics* 72, 31–45.
- Gerardi, Kristopher, 2017, GSE Mortgage Insurance Pricing, *Working paper* .
- Gerardi, Kristopher, Franklin Qian, and David Zhang, 2024, Mortgage lock-in, lifecycle migration, and the welfare effects of housing market liquidity .
- Gerardi, Kristopher, Franklin Qian, and David Zhang, 2026, Mortgage lock-in: a review of the literature .
- Gete, Pedro, Athena Tsouderou, and Susan M Wachter, 2023, Pricing mortgage stress: Evidence and policies .
- Gete, Pedro, and Franco Zecchetto, 2018, Distributional implications of government guarantees in mortgage markets, *The Review of Financial Studies* 31, 1064–1097.
- Glaeser, Edward L, Joshua D Gottlieb, and Joseph Gyourko, 2012, Can cheap credit explain the housing boom?, in *Housing and the financial crisis*, 301–359 (University of Chicago Press).
- Greenwald, Daniel, 2018, The mortgage credit channel of macroeconomic transmission .
- Greenwald, Daniel L, and Adam Guren, 2025, Do credit conditions move house prices?, *American Economic Review* 115, 3559–3596.
- Grundl, Serafin, and You Suk Kim, 2021, The marginal effect of government mortgage guarantees on homeownership, *Journal of Monetary Economics* 119, 75–89.
- Guren, Adam M, and Timothy J McQuade, 2020, How do foreclosures exacerbate housing downturns?, *The Review of Economic Studies* 87, 1331–1364.
- Hacamo, Isaac, Forthcoming, Interest rates and the distribution of house prices, *Review of*

- Head, Allen, Huw Lloyd-Ellis, and Hongfei Sun, 2014, Search, liquidity, and the dynamics of house prices and construction, *American Economic Review* 104, 1172–1210.
- Huh, Yesol, and You Suk Kim, 2022, The real effects of secondary market trading structure: Evidence from the mortgage market, *The Review of Financial Studies* 35, 3574–3616.
- Jeske, Karsten, Dirk Krueger, and Kurt Mitman, 2013, Housing, mortgage bailout guarantees and the macro economy, *Journal of Monetary Economics* 60, 917–935.
- Kalda, Ankit, Charles G Pearson, and David Sovich, 2025, Cost pass-through and mortgage credit: The case of guarantee fees, *Available at SSRN* .
- Keys, Benjamin J, Tanmoy Mukherjee, Amit Seru, and Vikrant Vig, 2010, Did securitization lead to lax screening? evidence from subprime loans, *The Quarterly journal of economics* 125, 307–362.
- Krainer, John, 2001, A theory of liquidity in residential real estate markets, *Journal of urban Economics* 49, 32–53.
- Landvoigt, Tim, Monika Piazzesi, and Martin Schneider, 2015, The housing market (s) of san diego, *American Economic Review* 105, 1371–1407.
- Lucas, Deborah, and Robert McDonald, 2010, Valuing government guarantees: Fannie and freddie revisited, in *Measuring and managing federal financial risk*, 131–154 (University of Chicago Press).
- Ngai, L Rachel, and Silvana Tenreyro, 2014, Hot and cold seasons in the housing market, *American Economic Review* 104, 3991–4026.
- Novy-Marx, Robert, 2009, Hot and cold markets, *Real Estate Economics* 37, 1–22.
- Ouazad, Amine, and Romain Rancière, 2019, Market frictions, arbitrage, and the capitalization of amenities, Technical report, National Bureau of Economic Research.
- Piazzesi, Monika, and Martin Schneider, 2009, Momentum traders in the housing market: Survey evidence and a search model, *American Economic Review* 99, 406–411.
- Piazzesi, Monika, Martin Schneider, and Johannes Stroebel, 2020, Segmented housing search, *American Economic Review* 110, 720–759.
- Poterba, James M, 1992, Taxation and housing: Old questions, new answers, *The American Economic Review* 82, 237–242.
- Sagi, Jacob S, 2021, Asset-level risk and return in real estate investments, *The Review of Financial Studies* 34, 3647–3694.
- Vickery, James I, and Joshua Wright, 2013, Tba trading and liquidity in the agency mbs market, *Economic Policy Review* 19.
- Wheaton, William C, 1990, Vacancy, search, and prices in a housing market matching model, *Journal of political Economy* 98, 1270–1292.

Wolf, Christian K, 2023, The missing intercept: A demand equivalence approach, *American Economic Review* 113, 2232–2269.

Zheng, Keling, 2024, Mortgage aggregation and credit supply .

Internet Appendix

A	Analysis on credit score manipulation	2
A.1	Presence of co-borrowers	2
A.2	Credit scores evolution prior to origination	3
A.2.1	Credit scores around cutoffs at the time of origination	3
A.2.2	Comparing CS at the initial credit check and underwriting CS	4
B	Data patterns around each credit score cutoff	7
C	Additional material for the effects on home purchases	10
D	Semi elasticity of relative home purchases by DTI	11
E	Additional material for regional variation in predicted home purchase effects	12
F	Additional material for further results	12
G	Additional material on financial intermediation	13
H	Computational Details of Housing Search Model	15
I	Additional Baseline Model Results	15
J	Decomposition of the Elasticity of Home Purchases to Mortgage Rates in Greenwald and Guren (2025)	16
J.1	Results	17
J.2	Proofs	18
K	Results with calibrated γ_0^S, γ_1^S	20

A Analysis on credit score manipulation

A.1 Presence of co-borrowers

If there is a co-borrower on a loan, the minimum CS among multiple borrowers on the same loan is used for underwriting. Thus, having a co-borrower would often decrease the underwriting CS, although it may increase the qualifying income. For this reason, the decision to have a co-borrower could be endogenous to the LLPA. For example, when the LLPA decreased more for CS 679 than CS 680, the downside of having a co-borrower with the CS below 680 has decreased, and more households may decide to include co-borrowers with lower CS on their loans. In that case, the DID regression would overstate the semi-elasticity for relative home purchases because the relative change in originations around the cutoff would be just driven by a downward shift in the underwriting CS because of more loans with co-borrowers.

We test whether the LLPA change led to changes in the presence of co-borrowers. Among the five credit score cutoffs in our sample, the LLPA declined more for CS right below the 680 and 720 cutoffs than CS right above each of the two cutoffs. Thus, including a co-borrower with CS below these cutoffs became less costly, and we may see more loans with co-borrowers with the underwriting CS below these cutoffs after the LLPA change. By contrast, in the rest of the credit score cutoffs (640, 760, and 780), the LLPA increased more for CS right below the cutoffs than CS right above the cutoffs. Thus, the hypothesis would predict fewer loans with co-borrowers for the underwriting CS below the cutoffs.

Table A.1 presents the regression results separately for the two types of CS cutoffs. We find no support for the hypothesis of a strategic decision to have a co-borrower.

Table A.1: Effects on whether a loan has a co-borrower

	subsample 1: CS cutoff $\in \{680, 720\}$			subsample 2: CS cutoff $\in \{640, 760, 780\}$		
	(1) Conventional conforming	(2) GSE	(3) Non-agency conforming	(4) Conventional conforming	(5) GSE	(6) Non-agency conforming
$1[t > Feb23] \times E[-\Delta LLPA]$	-0.002 (0.009)	0.010 (0.012)	-0.012 (0.019)	0.009 (0.008)	0.014 (0.010)	0.019 (0.012)
CS cutoff $\times$						
Income bin $\times$	Y	Y	Y	Y	Y	Y
Orig YM						
CS FE	Y	Y	Y	Y	Y	Y
County FE	Y	Y	Y	Y	Y	Y
N. Obs.	197,428	125,164	71,927	733,246	542,382	404,942
Adj. R^2	0.03	0.02	0.05	0.03	0.03	0.03

Note: this table shows loan-level regression in equation (1) for whether there is a co-borrower on a loan. Columns 1 to 3 use the subsample of CS cutoffs of 680 or 720, where the LLPA declined more for credit scores right below the cutoffs than those right above the cutoffs. Columns 4 to 6 use the subsample of CS cutoffs of 640, 760, or 780, where the LLPA increased more for credit scores right below the cutoffs than those right above the cutoffs.

A.2 Credit scores evolution prior to origination

In this section, we examine the possibility of changes in credit score manipulation in response to the LLPA change, using the CRISM data. The advantage of CRISM is that the data provides information about a history of different credit score types for the six months prior to mortgage origination. Thus, we can see if there is an unusual change in a borrower's CS history prior to origination in response to the LLPA change.

The downside of CRISM is that it lacks information about a borrower's income. Thus, it is difficult to identify whether a borrower's income is above the area median income, thereby figuring out whether the May LLPA change is applied to the borrower. Thus, we only consider borrowers with CS around the 680 and 720 cutoffs because the LLPA for borrowers with credit scores right below these cutoffs declined more than for borrowers with credit scores right above the same cutoffs in both the LLPA exemption for the LMI first-time home buyers and the LLPA change in May 2023 for other borrowers. In other words, borrowers with CS just below one of the two cutoffs (e.g., 670–679 or 710–719) will pay a lower LLPA than those with CS just above the cutoff (e.g., 680–689 or 720–729) in 2023, regardless of borrowers' incomes. In this context, our DID identification assumption will be violated if borrowers below the cutoffs are less likely to manipulate their credit scores above the cutoff in response to the LLPA change. We test this hypothesis in two ways.

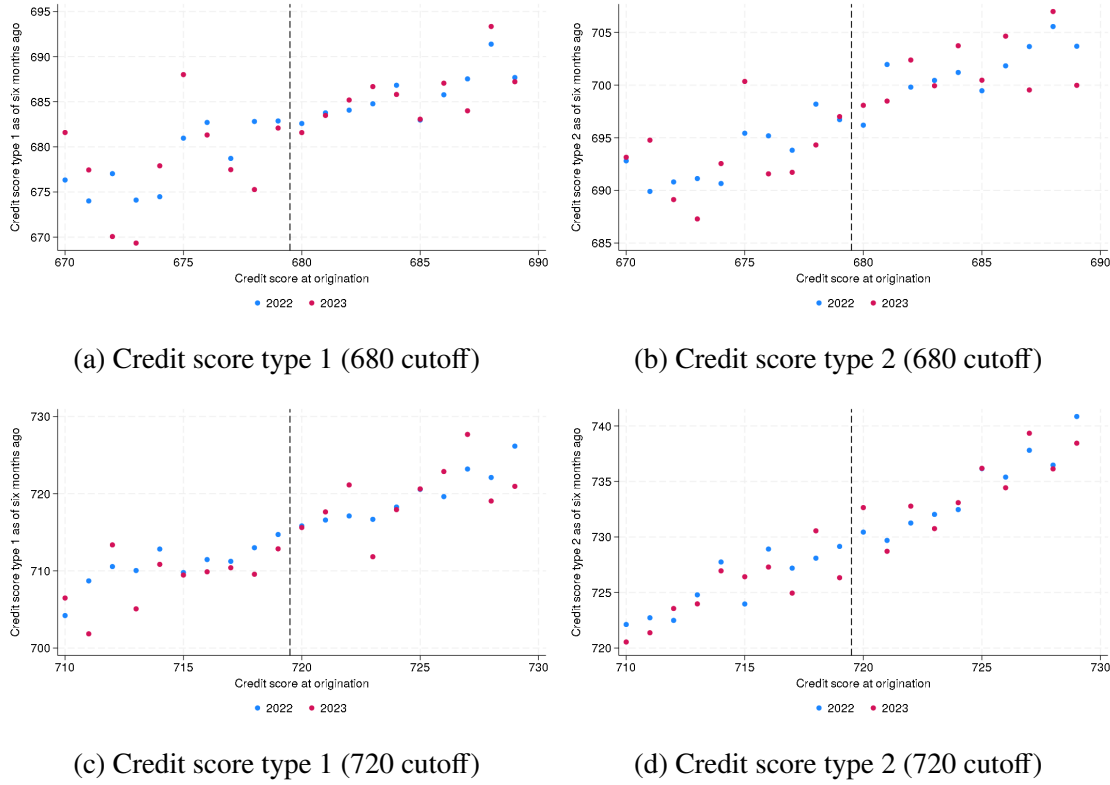
A.2.1 Credit scores around cutoffs at the time of origination

First, we test the manipulation hypothesis by examining CS as of six months prior to origination for borrowers with underwriting CS around one of the two cutoffs at the time of origination. If borrowers can manipulate their CS, it will be less costly (not only in terms of money but also time) to stop the manipulation once the credit score crosses a cutoff used for LLPA pricing. In our context, in 2022 when the new LLPA change became effective, borrowers may have incentives to manipulate their credit scores to be just over the 680 or 720 cutoff. In 2023, they will have smaller incentives to do the manipulation with the new LLPA pricing. Thus, the manipulation hypothesis would predict that, in 2022, borrowers with underwriting CS right above one of the two cutoffs are more likely to have a larger increase in their CS over the six months prior to the origination, compared with borrowers with origination credit scores right below the cutoffs. Then in 2023, there will be fewer borrowers with underwriting CS just above one of the two cutoffs, and there increases in CS above the cut-off should be smaller.

However, data patterns for GSE loan borrowers in Figure [A.1](#) are not consistent with the manipulation hypothesis. Aside some noises, six-month-ago CS are generally increasing in origination credit scores and smooth around the cutoffs both in 2022 and 2023. In addition, the relationship between six-month-ago CS and underwriting CS does not seem to have changed in 2023.

We also estimated the DID regression of two types of six-month-ago CS by comparing borrowers with underwriting CS around one of the two cutoffs in 2022 and 2023. The independent variable is whether the origination CS is below the cut-off. The manipulation hypothesis implies that those above the cut-off should have a lower increase in CS compared to those below the cutoff in 2023 as compared to 2022, which would imply a positive coefficient on $1[CS < \text{cutoff}] \times 1[t > \text{Jan2023}]$. The estimates presented in Table A.2 confirm that there was no statistically significant change in the relationship in 2023, with coefficients on being below the cut-off being negative which goes against the direction of a change in manipulation.

Figure A.1: Credit scores as of 6 months prior to origination



Note: residualized by origination year-month and zipcode

A.2.2 Comparing CS at the initial credit check and underwriting CS

The second way to test the manipulation hypothesis is to check whether borrowers who checked their CS prior to origination change their CS to be on the opposite side of CS cutoffs. That is, instead of using the underwriting CS as the independent variable, we use their CS at the time of their first credit inquiry. CRISM provides information about when a credit inquiry was made and the CS at the time of the inquiry.

We restrict the sample to borrowers who never had any credit check before they first appeared

Table A.2: Regressions with six-month-ago CS as the dependent variable

	Cutoff = 680		Cutoff = 720	
	(1) CS type 1	(2) CS type 2	(3) CS type 1	(4) CS type 2
$1[\text{CS} < \text{cutoff}] \times 1[t \geq \text{Jan2023}]$	-0.452 (1.960)	-0.309 (1.605)	-1.750 (1.675)	-0.519 (0.846)
Orig YM FE	Y	Y	Y	Y
Zip FE	Y	Y	Y	Y
N. Obs.	28,814	28,814	65,573	65,573
Adj. R^2	0.03	0.03	0.02	0.04
Note:				

in CRISM as of 6 months prior to origination. This sample selection excludes borrowers who potentially know their CS before appearing in CRISM and potentially engage in manipulation. Among borrowers who never had a credit check as of 6 months prior to origination, we identify CS at the time of the initial mortgage inquiry using two types of credit scores, which we call "initial CS."

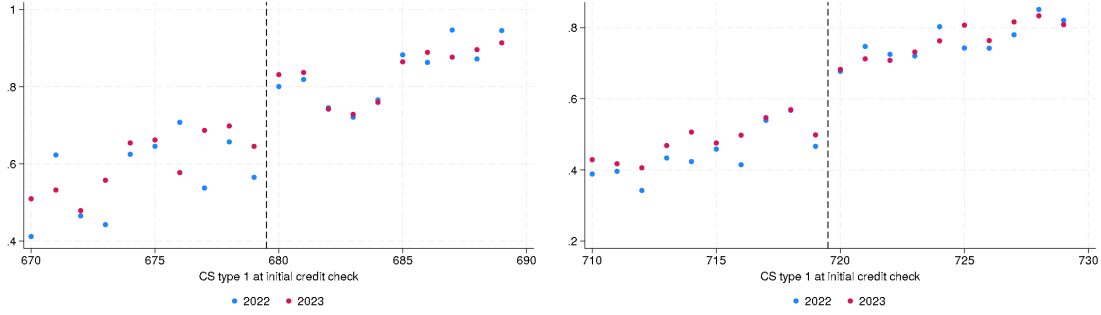
We note that as CRISM reports the initial CS from only one agency, whereas the tri-agency median credit score is used for origination. Furthermore, borrowers may correct their credit report after making the initial inquiry, thus changing their CS, and so some noise in the relationship between the initial check and the origination CS is expected. But, for a change in manipulation to explain our result, we would need the relationship between initial CS and origination CS to change over time.

More specifically, if there is a manipulation, where a borrower with the initial CS below a cutoff manipulates their credit scores, so their origination credit scores are located above the cutoff, this manipulation behavior will only violate our identification assumption if it changes over time. Empirically, this would imply that the share of borrowers with the origination credit score above the cutoff decreases in 2023 for borrowers with initial CS below the cutoff. This would imply a negative coefficient in front of $1[\text{Initial CS} < \text{cutoff}] \times 1[t > \text{Jan2023}]$.

Figure A.2 shows data patterns that are not consistent with the sophisticated manipulation behavior that would respond to the LLPA change. If anything, panels (a) and (b) show that credit scores are rather persistent over time instead of being subject to manipulation. The sophisticated manipulation would predict that those with initial CS below the cutoff would be less likely to have their origination credit scores above the cutoff after the LLPA change. The data do not seem to support the hypothesis.

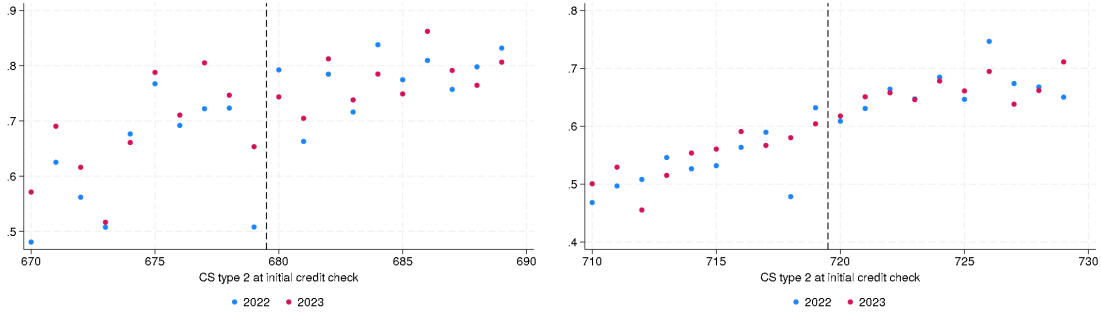
Table A.3 presents the estimates of the DID regression testing the hypothesis. The estimate in column 3 is marginally significant, but the coefficient estimate has the opposite sign to what a change in manipulation hypothesis predicts.

Figure A.2: Share of borrowers with origination credit scores above cutoff



(a) $1[\text{Origination CS} \geq 680]$;
(by credit score type 1 at initial check)

(b) $1[\text{Origination CS} \geq 720]$;
(by credit score type 1 at initial check)



(c) $1[\text{Origination CS} \geq 680]$;
(by credit score type 2 at initial check)

(d) $1[\text{Origination CS} \geq 720]$;
(by credit score type 2 at initial check)

Note: residualized by origination year-month and zipcode

Table A.3: Credit score manipulation (in terms of initial CS)

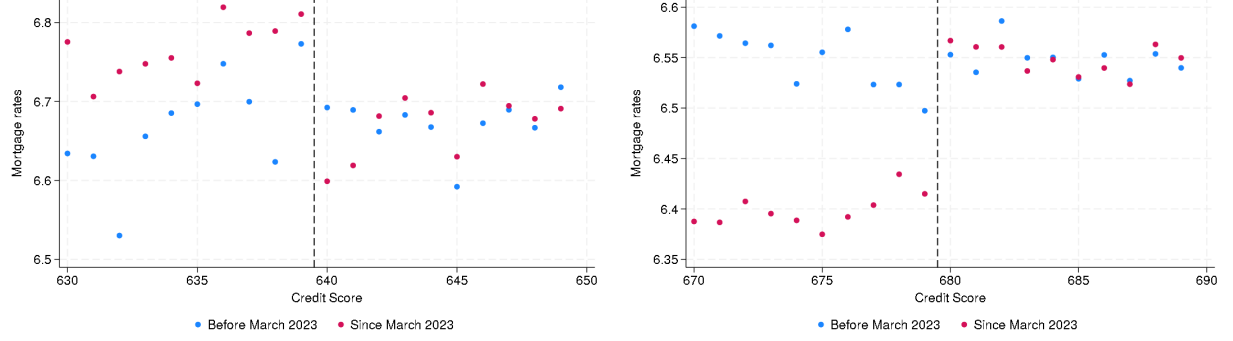
	Cutoff=680, Dep. Var. = $1[\text{Origination CS} \geq 680]$		Cutoff=720, Dep. Var. = $1[\text{Origination CS} \geq 720]$	
	(1) Initial CS type 1 $\in [670, 689]$	(2) Initial CS type 2 $\in [670, 689]$	(3) Initial CS type 1 $\in [710, 729]$	(4) Initial CS type 2 $\in [710, 729]$
$1[\text{Initial CS} < \text{cutoff}] \times \text{Post}$	0.039 (0.059)	0.049 (0.037)	0.038* (0.020)	0.013 (0.021)
Orig YM FE	Y	Y	Y	Y
Zip FE	Y	Y	Y	Y
N. Obs.	3,080	2,310	9,627	8,897
Adj. R^2	0.13	0.01	0.13	0.02

Note:

B Data patterns around each credit score cutoff

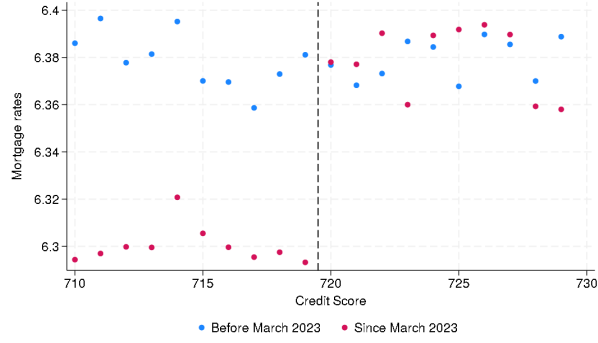
Figures [A.3](#) and [A.4](#) presents data patterns driving the estimates graphically after residualizing mortgage rates for GSE loans by LTVs, the lender $\times$ county fixed effects, and the time fixed effects. In all panels, there is a visibly clear change at the cutoff before and after the LLPA change. Moreover, mortgage rates changed in a way that is consistent with relative changes in the LLPA around each credit score cutoff. For example, borrowers with credit scores below 680 saw a decline in mortgage rates relative to those on the other side of the 680 cutoff because the former group had a larger decline in the LLPAs. Borrowers with credit scores below the 760 cutoff saw an increase in mortgage rates relative to those on the other side of the 760 cutoff because the former group had a smaller increase in the LLPAs. We also report figures for net origination charges, which also show similar patterns.

Figure A.3: Mortgage rates of GSE loans

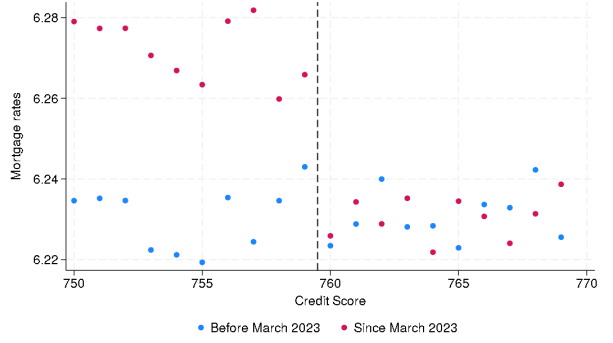


(a) credit score cutoff 640

(b) credit score cutoff 680



(c) credit score cutoff 720



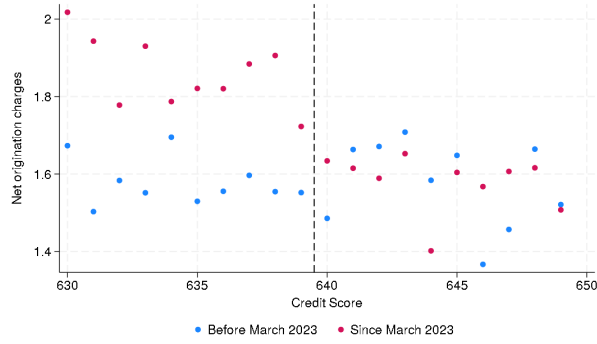
(d) credit score cutoff 760



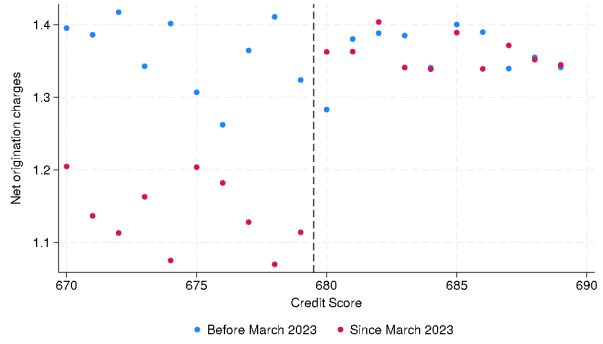
(e) credit score cutoff 780

Note: this figure presents mortgage rates of GSE loans for each credit score point, residualized by income bin-by-origination year-month fixed effects and lender-by-county fixed effects.

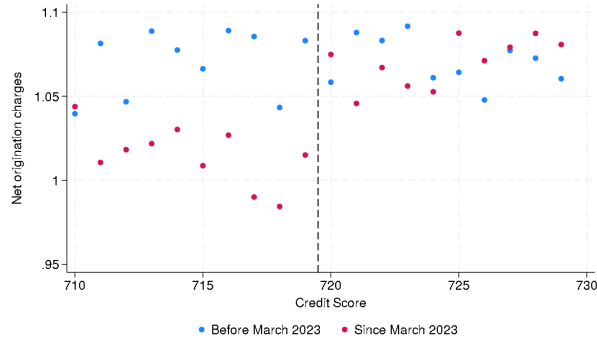
Figure A.4: Net origination charges of GSE loans (% of loan amounts)



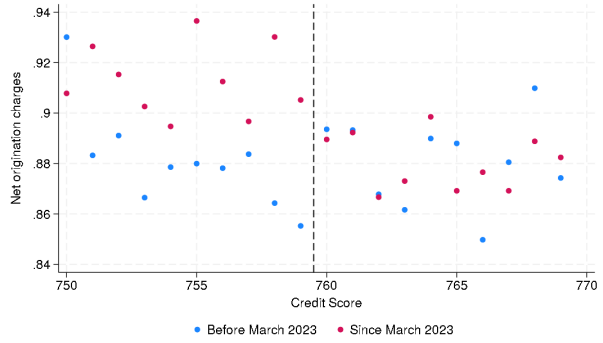
(a) credit score cutoff 640



(b) credit score cutoff 680



(c) credit score cutoff 720



(d) credit score cutoff 760

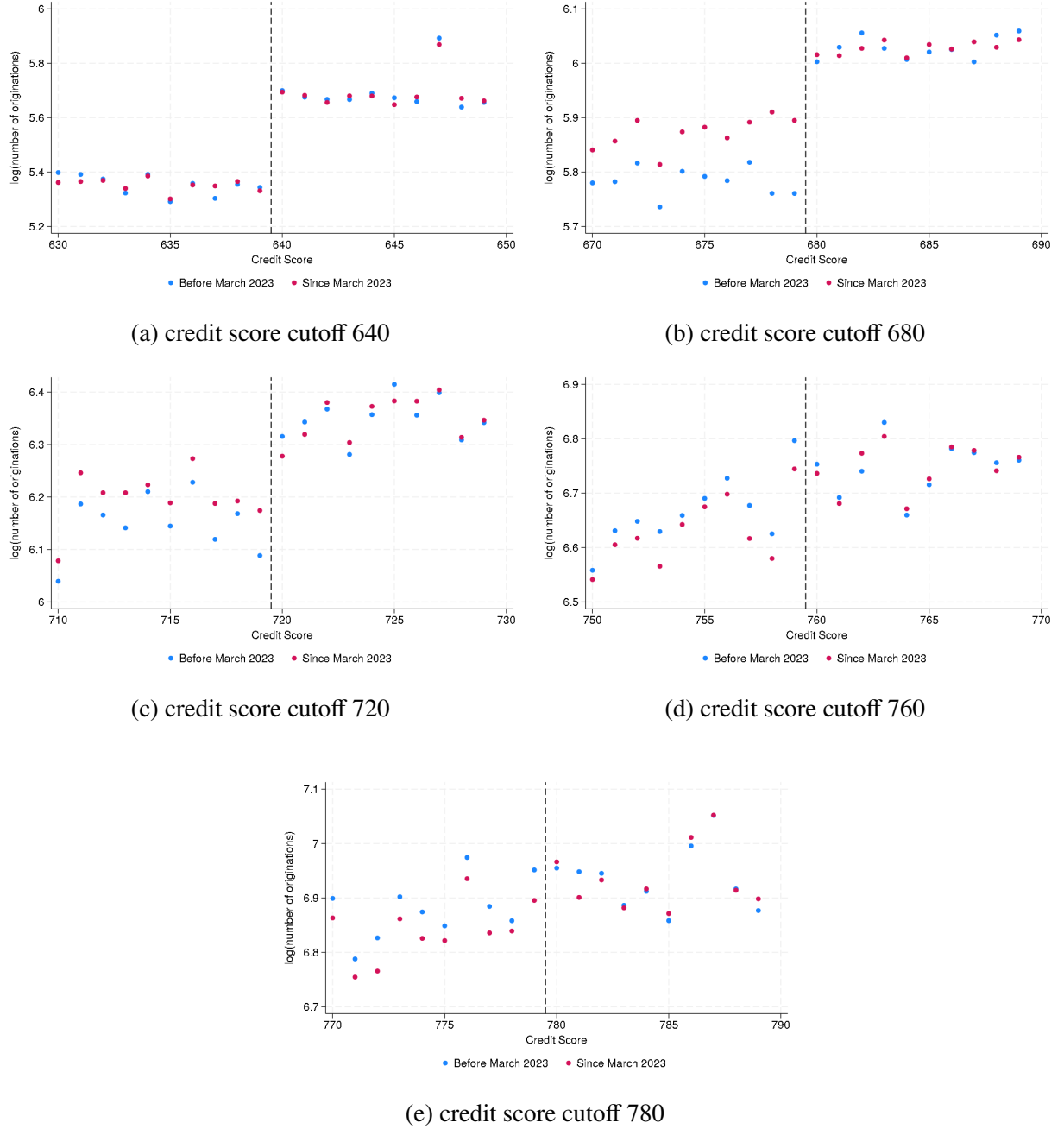


(e) credit score cutoff 780

Note: this figure presents net origination charges of GSE loans for each credit score point, residualized by income bin-by-origination year-month fixed effects and lender-by-county fixed effects.

C Additional material for the effects on home purchases

Figure A.5: Log of total number of home-purchase loan originations



Note: this figure presents log of total home-purchase originations for each credit score point, residualized by origination year-month fixed effects.

D Semi elasticity of relative home purchases by DTI

We examine whether the semi-elasticity of relative home purchase volume varies with DTI, in light of findings that the binding DTI constraint is the main mechanism through which interest rates influence home purchases (Bhutta and Ringo, 2021; Bosshardt et al., 2024a). Because the actual DTI is affected by the LLPA change directly, we construct the counterfactual DTI (“cDTI”, henceforth) to compare borrowers that paid different LLPAs.

Specifically, our construction of the cDTI involves analyzing two groups of credit scores, depending on whether their average LLPA ($E[\Delta LLPA]$) increased or decreased. For credit scores with $E[\Delta LLPA] < 0$, we reverse this reduction on actual DTIs of loans originated *after* the LLPA reduction. This allows us to determine what their DTIs would have been if the LLPA had remained at its previous, higher level, and to identify potentially binding constraints. For credit scores with $E[\Delta LLPA] > 0$, we apply this increase to calculate cDTIs of loans originated *before* the increase. This allows us to identify borrowers whose DTIs would have been binding if they had originated loans after the increase in the LLPA.

Moreover, we also take into account the effects of net discount points purchased (points purchased net of lender credits) on the actual DTI in constructing the cDTI. Because of the high interest rate environment during our sample period, a majority of borrowers had some discount points purchased. Thus, to make cDTIs comparable across borrowers with different points purchased, the cDTI is constructed assuming no points purchased by undoing the effects of the actual points purchased on interest rates.

Defining $PI(r_{it})$ to be the principal and interest payment with interest rate r_{it} of borrower i originating a loan in time t , the cDTI is calculated as follows:

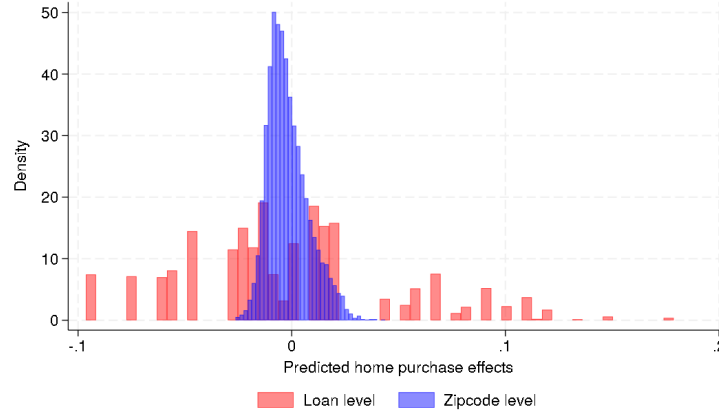
$$cDTI_{it} = DTI_{it} + \frac{PI(\tilde{r}_{it}) - PI(r_{it})}{\text{income}}, \quad (22)$$

where r_{it} and $\tilde{r}_{it}$ refer to the actual and counterfactual interest rates. The latter is defined as follows:

$$\tilde{r}_{it} = \begin{cases} r_{it} + \frac{\text{NetPoints} - E[\Delta LLPA]}{3}, & \text{if } E[\Delta LLPA] < 0 \text{ and } t > \text{Feb23} \\ r_{it} + \frac{\text{NetPoints} + E[\Delta LLPA]}{3}, & \text{if } E[\Delta LLPA] > 0 \text{ and } t \leq \text{Feb23} \\ r_{it} + \frac{\text{NetPoints}}{3}, & \text{otherwise.} \end{cases} \quad (23)$$

E Additional material for regional variation in predicted home purchase effects

Figure A.6: Predicted home purchase effects: loan level vs zipcode level



F Additional material for further results

Table A.4: Heterogeneous effects on the number of home-purchase loan originations by CS and income

	(1)	(2)	(3)
	CS < 700	CS ∈ [700, 740)	CS ≥ 740
$1[t > Feb23] \times E[-\Delta LPA]$	0.095*** (0.019)	0.177*** (0.054)	0.171*** (0.043)
$1[t > Feb23] \times E[-\Delta LPA] \times 1[\frac{Income}{AMI} \in (P_{33}, P_{66})]$	0.054** (0.026)	0.095* (0.053)	-0.029 (0.058)
$1[t > Feb23] \times E[-\Delta LPA] \times 1[\frac{Income}{AMI} \in (P_{66}, P_{90})]$	0.068** (0.034)	-0.045 (0.108)	0.064 (0.065)
$1[t > Feb23] \times E[-\Delta LPA] \times 1[\frac{Income}{AMI} > P_{90}]$	0.149** (0.068)	-0.093 (0.130)	-0.165*** (0.052)
Cutoff × Income bin × Orig YM	Y	Y	Y
CS FE	Y	Y	Y
N. Obs.	1,440	720	1,440
Pseudo R^2	0.92	0.89	0.89

Note: this table presents Poisson regression estimates of equation (2) for relative home purchase volumes. P_n refers to the n-th percentile of $\frac{Income}{AMI}$. Standard errors are clustered by credit score.

Table A.5: Effects on loan amounts and home prices

	log(House price)					log(Loan amount)				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	All loans	GSE	Non-agency conforming	Jumbo	FHA/VA	All loan	GSE	Non-agency conforming	Jumbo	FHA/VA
$1[t > Feb23] \times E[-\Delta LPA]$	0.005 (0.003)	0.019*** (0.007)	-0.007 (0.008)	0.003 (0.017)	0.005* (0.003)	0.004 (0.003)	0.028*** (0.007)	-0.003 (0.008)	0.010 (0.014)	0.006* (0.003)
CS cutoff $\times$										
Income bin $\times$	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
Orig YM										
CS FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
County FE	Y	Y	Y	Y	Y	Y	Y	Y	Y	Y
N. Obs.	1,073,021	461,844	269,059	60,953	280,239	1,073,567	462,932	269,850	59,888	279,980
Adj. R^2	0.63	0.52	0.59	0.43	0.62	0.55	0.47	0.53	0.48	0.60

Note: this table presents loan-level regression estimates of equation (1) for log of the price of the house purchased and log of loan amount. Standard errors are clustered by credit score.

G Additional material on financial intermediation

Table A.6: Effects on borrowing costs by bank vs nonbank

	Mortgage Rate			Net Orig Charges		
	(1) GSE	(2) Non-agency conforming	(3) Conventional conforming	(4) GSE	(5) Non-agency conforming	(6) Conventional conforming
$1[t > Feb23]$ $\times E[-\Delta LPA]$	-0.187*** (0.028)	-0.014 (0.024)	-0.082*** (0.020)	-0.259*** (0.038)	-0.032 (0.023)	-0.118*** (0.018)
Cutoff $\times$ Income bin $\times$ Orig YM	Y	Y	Y	Y	Y	Y
CS FE	Y	Y	Y	Y	Y	Y
LLPA grid FE	Y	Y	Y	Y	Y	Y
Lender $\times$ County FE	Y	Y	Y	Y	Y	Y
N. Obs.	92,497	115,448	237,750	91,536	113,670	234,709
Adj. R^2	0.71	0.77	0.75	0.26	0.45	0.35
(a) Banks						
	Mortgage Rate			Net Orig Charges		
	(1) GSE	(2) Non-agency conforming	(3) Conventional conforming	(4) GSE	(5) Non-agency conforming	(6) Conventional conforming
$1[t > Feb23]$ $\times E[-\Delta LPA]$	-0.221*** (0.014)	-0.196*** (0.045)	-0.208*** (0.012)	-0.297*** (0.023)	-0.401*** (0.064)	-0.313*** (0.020)
Cutoff $\times$ Income bin $\times$ Orig YM	Y	Y	Y	Y	Y	Y
CS FE	Y	Y	Y	Y	Y	Y
LLPA grid FE	Y	Y	Y	Y	Y	Y
Lender $\times$ County FE	Y	Y	Y	Y	Y	Y
N. Obs.	284,009	56,967	388,039	279,924	54,916	380,893
Adj. R^2	0.67	0.71	0.66	0.27	0.37	0.28
(b) Nonbanks						

Note: this table presents Poisson regression estimates of equation (2) for number of home purchases originations. We measure the income by the median point of incomes relative to the area median income in each income bin. We subtracted the median income relative to the area median income in these regressions. We also subtracted 740 from credit scores and divided it by 100. Standard errors are clustered by credit score.

Table A.7: Effects on nonbank share of originations

	(1) All loans	(2) Conventional conforming	(3) GSE	(4) Non-agency conforming
$1[t > Feb23] \times E[-\Delta LPA]$	0.019*** (0.005)	0.035*** (0.008)	-0.002 (0.008)	0.048*** (0.014)
CS cutoff $\times$ Income bin $\times$ Orig YM	Y	Y	Y	Y
CS FE	Y	Y	Y	Y
County FE	Y	Y	Y	Y
N. Obs.	1,088,460	732,540	442,874	223,766
Adj. R^2	0.13	0.11	0.10	0.13

Note: this table presents loan-level regression estimates of equation (1) for the nonbank share of originations by loan type. Standard errors are clustered by credit score.

H Computational Details of Housing Search Model

We first assume M^S, M^B, V^B, V^S , then solve for the matching rates λ^S, λ^B using Equation (16), then update $G(r)$ using flow equality using Equation (17), then invert to find trade cut-offs $\bar{\epsilon}(r_i)$, then updating V^B, V^S via Bellman equations, then re-solving for trade-cutoffs $\bar{\epsilon}(r_i)$ in Equation (15), then updating M^B using Equation (17), until convergence. Instead of fully updating to the new implied values each iteration, a dampening parameter of 0.3 where each new value is equal to (1-0.3) times the older value plus 0.3 times the new value is helpful for achieving convergence across many runs.

Then, we use an outer loop that conducts a bisection search over M^S to solve for a desired μ^S as a model primitive, where μ^S is calibrated to the buyer success probability s and must therefore be equal to $\sum_i \mu^B(r_i)s$. Following that, we use a second outer loop incorporating the first outer loop to solve for the inflow parameters $\gamma_0^B, \gamma_c^b, \gamma_k^b, \gamma_{ck}^b, \xi_{cklfb(j)}$, the latter of which is via nested fixed point following BLP (Berry et al., 1995) to fit the market shares in each credit score by income by LTV by FHA type by bank type cell. Counterfactual simulations involve going through the same two-outer-loop procedure, but not updating the $\xi_{cklfb(j)}$ in the second outer loop. We construct model counterparts to the empirical May 2023 relative home purchase elasticities to g-fee changes by comparing borrowers around each CS bin above and below the cut-offs, computing the relative home purchase semi-elasticities within each of the five cut-offs we use in the main empirical analysis, and equal weighting them as we do in the main empirical analysis.

I Additional Baseline Model Results

Table A.8: Effect of Uniform G-Fee Increase (+25 bps) on Home Purchases by Income

Income Group	Δ Purchases (percentage point)	
	Fixed Supply	Free Entry
LMI	+0.11	-2.43
g_1 (low)	+0.08	-2.50
g_2	-0.12	-2.83
g_3	-0.28	-3.11
g_4 (high)	-0.35	-3.21
Total	-0.00	-2.63

Notes: Change in total home purchases under a uniform 25 basis point g-fee increase relative to the post-May 2023 LLPA schedule, by income group. Fixed Supply holds seller mass constant; Free Entry allows sellers to enter proportionally with buyers.

Table A.9: Effect of Uniform G-Fee Increase (+25 bps) on Home Purchases by Income in a Higher-DOM Market (DOM = 64)

Income Group	Δ Purchases (percentage point)	
	Fixed Supply	Free Entry
LMI	+0.06	−2.95
g_1 (low)	+0.05	−3.04
g_2	−0.06	−3.43
g_3	−0.15	−3.76
g_4 (high)	−0.19	−3.87
Total	−0.00	−3.19

Notes: Change in total home purchases under a uniform 25 basis point g-fee increase relative to the post-May 2023 LLPA schedule, by income group, in a market recalibrated to an average days-on-market of 64 days. Fixed Supply holds seller mass constant; Free Entry allows sellers to enter proportionally with buyers.

Table A.10: Effect of May 2023 g-fee change on Home Purchases by Income

Income Group	Δ Purchases (percentage point)	
	Fixed Supply	Free Entry
LMI	+0.07	+0.27
g_1 (low)	+0.03	+0.24
g_2	−0.07	+0.14
g_3	−0.10	+0.11
g_4 (high)	−0.19	+0.02
Total	+0.00	+0.21

Notes: Change in total home purchases across all channels following the 2023 LLPA reform, by income group. Fixed Supply holds seller mass constant; Free Entry allows sellers to enter proportionally with buyers.

J Decomposition of the Elasticity of Home Purchases to Mortgage Rates in [Greenwald and Guren \(2025\)](#)

This section decomposes the partial equilibrium effect of mortgage rates on homeownership in the framework of [Greenwald and Guren \(2025\)](#) into two channels: a *user cost channel*, through which higher rates reduce the value of mortgage borrowing to households that are impatient relative to lenders, and a *PTI channel*, through which higher rates tighten payment-to-income

constraints.¹ We show that the relative contribution of each channel are similar.

J.1 Results

We first introduce the objects that appear in the decomposition. In the model of [Greenwald and Guren \(2025\)](#), the marginal collateral value

$$C_B = \mu_B \cdot F^{LTV} \cdot \theta^{LTV} \quad (24)$$

captures the shadow value to borrowers of housing as collateral ([Greenwald and Guren, 2025](#), footnote 23). Here θ^{LTV} is the maximum loan-to-value ratio, F^{LTV} is the fraction of borrowers whose LTV constraint binds (rather than the PTI constraint), and $\mu_B = 1 - \Omega_M^B - r_B^* \Omega_X^B$ is the per-dollar surplus from borrowing, where Ω_M^B and Ω_X^B are the borrower's present values of principal and coupon streams following the mortgage technology of [Greenwald \(2018\)](#). Here, Ω_M^B and Ω_X^B depend on the borrower's discount factor β_B but not on the mortgage rate r_B^* . A change in r_B^* affects C_B through μ_B and through F^{LTV} , and the latter shift away from LTV constrained borrowers (rather than PTI constrained borrowers) can be characterized as the PTI channel.

Proposition J.1 (Decomposition of the PE Homeownership Elasticity). *Fix house prices p and rents q at their equilibrium values. The partial equilibrium (PE) response of the borrower homeownership rate HOR_B to the mortgage rate r_B^* is*

$$\frac{d HOR_B}{d r_B^*} = -\Lambda \left(\underbrace{\Omega_X^B \cdot F^{LTV}}_{\text{user cost}} + \underbrace{\mu_B \cdot \left| \frac{dF^{LTV}}{d r_B^*} \right|}_{\text{PTI}} \right), \quad (25)$$

where $\Lambda > 0$ is a positive constant that does not depend on r_B^* . The derivative of F^{LTV} with respect to r_B^* is

$$\frac{dF^{LTV}}{d r_B^*} = -\frac{\phi(z_{\bar{e}}/\sqrt{2})}{\sqrt{2} \sigma_e (r_B^* + \nu + \alpha)}, \quad (26)$$

where $\bar{e} \equiv \theta^{LTV} p H^* / [(\theta^{PTI} - \kappa) y_B / (r_B^* + \nu + \alpha)]$ is the income threshold between the LTV and PTI constraints from equations (3)–(4) of [Greenwald and Guren \(2025\)](#), $z_{\bar{e}} = (\log \bar{e} + \frac{1}{2} \sigma_e^2) / \sigma_e$, and ϕ is the standard normal density. The relative contribution of each channel is

$$\text{User cost share} = \frac{\Omega_X^B \cdot F^{LTV}}{\Omega_X^B \cdot F^{LTV} + \mu_B \cdot \left| \frac{dF^{LTV}}{d r_B^*} \right|}, \quad \text{PTI share} = 1 - \text{User cost share}. \quad (27)$$

¹In our paper, we equivalently refer to this as the debt-to-income, or DTI, constraint. We use the PTI terminology in this Appendix section to be consistent with the terminology in [Greenwald and Guren \(2025\)](#).

Proof. See Section J.2. □

Here σ_e is the dispersion of idiosyncratic income shocks, ν is the mortgage amortization rate, α and κ are PTI offsets for taxes/insurance and non-housing debt, θ^{PTI} is the maximum payment-to-income ratio, y_B is borrower income, and H^* is the new house size; all are defined in Greenwald and Guren (2025) and summarized in their Table 1.

Table A.11 reports the decomposition evaluated at their baseline calibration. For a small increase in interest rate, the two channels contribute comparably to the ownership decline, with the user cost channel being slightly larger.

Table A.11: Partial Equilibrium Decomposition of Homeownership Response to Mortgage Rates in Greenwald and Guren (2025)

Channel	Share
User cost (via μ_B)	56.5%
PTI (via F^{LTV})	43.5%

Notes: Shares are computed from equation (27) evaluated at steady state. “User cost” is the share attributable to $d\mu_B/dr_B^* = -\Omega_X^B$: a higher mortgage rate reduces the surplus from the wedge between borrower and lender discount rates. “PTI” is the share attributable to dF^{LTV}/dr_B^* : a higher rate tightens payment-to-income constraints, reducing the fraction of borrowers who can leverage housing collateral.

J.2 Proofs

The proof of Proposition J.1 uses two intermediate results, which we state and prove first.

Definition J.2 (User Cost Rate). The *user cost rate* is

$$\mathcal{UC}(C_B) \equiv \frac{1 - C_B}{\beta_B} - 1 + \delta + (1 - \rho) C_B. \quad (28)$$

Lemma J.3 (Steady-State Ownership Threshold). *In steady state, the marginal borrower’s ownership surplus $\bar{\omega}_B$ satisfies*

$$\bar{\omega}_B = \frac{p}{q} \mathcal{UC}(C_B) - 1. \quad (29)$$

Proof of Lemma J.3. Evaluate the borrower’s demand curve (Greenwald and Guren, 2025, equation 5) in steady state:

$$p = \frac{\beta_B [(1 + \bar{\omega}_B) q + (1 - \delta - (1 - \rho) C_B) p]}{1 - C_B}.$$

Multiplying both sides by $(1 - C_B)/\beta_B$ and rearranging:

$$p \left[\frac{1 - C_B}{\beta_B} - 1 + \delta + (1 - \rho) C_B \right] = (1 + \bar{\omega}_B) q.$$

The bracketed expression is $\mathcal{UC}(C_B)$ from Definition J.2. Dividing by q and subtracting 1 yields (29). $\square$

Proof of Proposition J.1. The proof proceeds in three steps.

Step 1: From r_B^ to $\bar{\omega}_B$.* By Lemma J.3, $\bar{\omega}_B$ depends on r_B^* only through C_B . With p and q fixed,

$$\frac{d\bar{\omega}_B}{dr_B^*} = \frac{p}{q} \frac{d\mathcal{UC}}{dC_B} \frac{dC_B}{dr_B^*}. \quad (30)$$

From (28), $d\mathcal{UC}/dC_B = -1/\beta_B + (1 - \rho) < 0$, since $1/\beta_B > 1$.

Step 2: From $\bar{\omega}_B$ to HOR_B . The homeownership rate satisfies $HOR_B = 1 - \Gamma_{\omega,B}(\bar{\omega}_B)$ (Greenwald and Guren, 2025, p. 3581), so

$$\frac{dHOR_B}{dr_B^*} = -\gamma_{\omega,B}(\bar{\omega}_B) \frac{d\bar{\omega}_B}{dr_B^*}.$$

Substituting (30):

$$\frac{dHOR_B}{dr_B^*} = -\gamma_{\omega,B}(\bar{\omega}_B) \frac{p}{q} \frac{d\mathcal{UC}}{dC_B} \frac{dC_B}{dr_B^*}. \quad (31)$$

Defining $\Lambda \equiv \gamma_{\omega,B}(\bar{\omega}_B) (p/q) (-d\mathcal{UC}/dC_B) \theta^{LTV} > 0$, this becomes $dHOR_B/dr_B^* = (\Lambda/\theta^{LTV}) dC_B/dr_B^*$. It remains to calculate dC_B/dr_B^* .

Step 3: Calculating dC_B/dr_B^ .* Since $C_B = \mu_B F^{LTV} \theta^{LTV}$,

$$\frac{dC_B}{dr_B^*} = \theta^{LTV} \left[F^{LTV} \frac{d\mu_B}{dr_B^*} + \mu_B \frac{dF^{LTV}}{dr_B^*} \right].$$

We show that both terms are negative. For the first, $\mu_B = 1 - \Omega_M^B - r_B^* \Omega_X^B$ and Ω_M^B, Ω_X^B are invariant to r_B^* ,² so $d\mu_B/dr_B^* = -\Omega_X^B < 0$.

For the second, $F^{LTV} = 1 - \Phi(z_{\bar{e}}/\sqrt{2})$, where Φ is the standard normal CDF and $\bar{e}$ is the ratio of the LTV limit to the PTI limit from equations (3)–(4) of Greenwald and Guren (2025). Since the PTI denominator is $r_B^* + \nu + \alpha$, we have $\log \bar{e} = \text{const} + \log(r_B^* + \nu + \alpha)$, so

$$\frac{d \log \bar{e}}{dr_B^*} = \frac{1}{r_B^* + \nu + \alpha}.$$

² Ω_M^B and Ω_X^B are the borrower-side analogues of the saver bond prices Q_M and Q_X . They depend on β_B, ν, ρ , and π , but not on r_B^* .

Differentiating $F^{LTV} = 1 - \Phi(z_{\bar{e}}/\sqrt{2})$ and applying $z_{\bar{e}} = (\log \bar{e} + \frac{1}{2}\sigma_e^2)/\sigma_e$:

$$\frac{dF^{LTV}}{dr_B^*} = -\frac{1}{\sqrt{2}} \phi\left(\frac{z_{\bar{e}}}{\sqrt{2}}\right) \frac{dz_{\bar{e}}}{dr_B^*} = -\frac{\phi(z_{\bar{e}}/\sqrt{2})}{\sqrt{2} \sigma_e (r_B^* + \nu + \alpha)} < 0,$$

which is (26). Since both terms are negative,

$$\frac{dC_B}{dr_B^*} = -\theta^{LTV} \left(\Omega_X^B \cdot F^{LTV} + \mu_B \cdot \left| \frac{dF^{LTV}}{dr_B^*} \right| \right).$$

Substituting into $dHOR_B/dr_B^* = (\Lambda/\theta^{LTV}) dC_B/dr_B^*$ yields (25). □

K Results with calibrated γ_0^S, γ_1^S

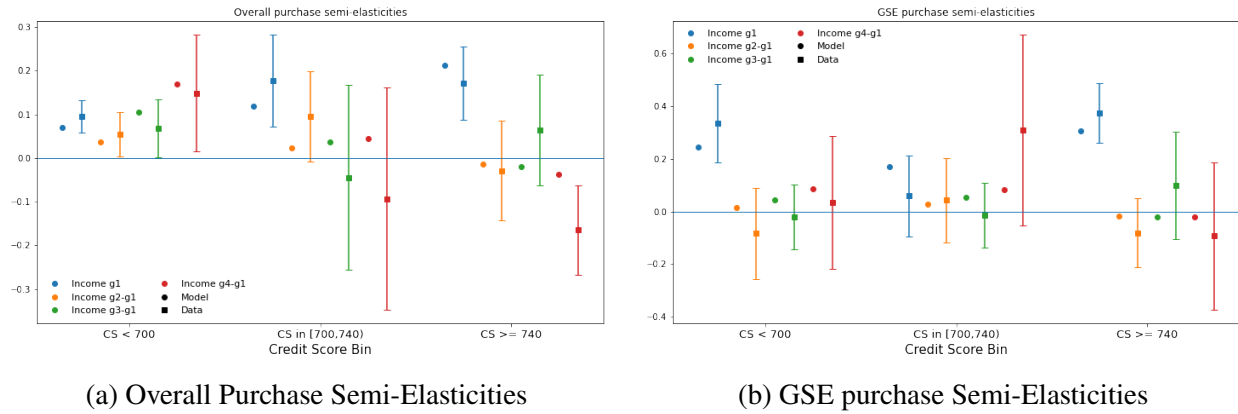
Table A.12: Calibrated model parameters with calibrated γ_0^S, γ_1^S

Parameter	Description	Value
κ	Buyer exit rate	0.3077
δ^S	Seller discount rate increment	1.1277
ψ	Mortgage rate weight in flow payoff	13.7366
ρ^b	Buyer flow payoff from owning	68.1108
σ	Match shock std. dev.	14.0094
ϕ	Matching function elasticity (buyers)	0.7593
γ_1^S	Seller entry sensitivity to V^S	0.0123
γ_0^B	Rate sensitivity (multiplier)	0.0384
γ_c^b	Rate sensitivity (credit score gradient)	3.8955
γ_k^b	Rate sensitivity (income gradient)	0.6363
γ_{ck}^b	Rate sensitivity (credit score by income gradient)	-1.1624

Table A.13: Comparison of model output to targeted moments, calibrated γ_0^S, γ_1^S

Statistic	Model Output	Target	Difference
Buyer weeks to success	10.705	12.000	-1.295
Seller days on market (DOM)	54.061	52.000	2.061
Expected number of offers before success	2.426	2.500	-0.074
Average house price (\$10,000s)	44.361	43.295	1.066
Std. dev. of log price	0.096	0.093	0.003
Relative home purchase semi-elasticity (100 bps g-fee)	0.148	0.139	0.009
Non-bank share change (100 bps g-fee)	0.013	0.019	-0.006
Aggregate home sales growth (100 bps rate increase)	0.132	0.134	-0.002

Figure A.7: Fit of our model to overall and GSE purchase semi-elasticities by income and CS, calibrated γ_0^S, γ_1^S



Note: Round dots represent model predictions, whereas square dots with error bars represent estimates from Table A.4 in Figure A.7(a) and analogous estimates for GSE loans only in Figure A.7(b).

Table A.14: Effect of Uniform G-Fee Increase (+25 bps) on GSE Markups and Profits, Calibrated γ_0^S, γ_1^S

FICO	$\Delta \text{Purch.}\%$	$\Delta \text{GSE}\%$	Markup (bps)			Profit (\$B)		$\Delta \pi$ (\$B)
			Old	New	Δ	Old	New	
780	-2.80	-6.8	56	81	+25	1.14	1.54	+0.40
760	-1.47	-4.1	51	76	+25	0.56	0.80	+0.24
740	-0.86	-2.9	51	76	+25	0.47	0.68	+0.21
720	-0.55	-2.4	45	70	+25	0.33	0.50	+0.17
700	-0.15	-2.0	41	66	+25	0.23	0.37	+0.13
680	+0.32	-2.0	51	76	+25	0.20	0.29	+0.09
660	+1.00	-1.9	51	76	+25	0.11	0.16	+0.05
640	+1.44	-1.9	53	78	+25	0.06	0.09	+0.03
620	+2.11	-2.2	83	108	+25	0.06	0.07	+0.02
Total	-0.87	-4.1	51	76	+25	3.17	4.51	+1.34

Notes: This table reports outcomes under a uniform 25 basis point increase in upfront g-fees across all FICO-LTV cells. Markups are LLPA minus model-implied cost, in basis points. Profits are markup $\times$ GSE origination volume, scaled to \$616B total GSE originations. Old markups and profits correspond to the post-May 2023 LLPA schedule. Seller entry follows the logit specification in Equation (6), calibrated to the aggregate interest elasticity of [Hacamo \(Forthcoming\)](#).

Table A.15: Effect of Uniform G-Fee Increase (+25 bps) on Home Purchases by Income, Calibrated γ_0^S, γ_1^S

Income Group	Δ Purchases (percentage point)
LMI	−0.73
g_1 (low)	−0.77
g_2	−1.01
g_3	−1.21
g_4 (high)	−1.29
Total	−0.87

Notes: Change in total home purchases under a uniform 25 basis point g-fee increase relative to the post-May 2023 LLPA schedule, by income group, under logit seller entry calibrated to the aggregate interest elasticity of [Hacamo \(Forthcoming\)](#).

Table A.16: Effect of Uniform G-Fee Increase (+25 bps) in a Higher-DOM Market (DOM = 64), Calibrated γ_0^S, γ_1^S

FICO	Δ Purch. %	Δ GSE %	Markup (bps)			Profit (\$B)		$\Delta\pi$ (\$B)
			Old	New	Δ	Old	New	
780	−1.67	−6.3	55	80	+25	1.49	2.02	+0.53
760	−1.01	−3.8	51	76	+25	0.55	0.79	+0.24
740	−0.64	−2.5	51	76	+25	0.40	0.58	+0.18
720	−0.45	−2.0	46	71	+25	0.27	0.41	+0.14
700	−0.19	−1.6	41	66	+25	0.18	0.29	+0.11
680	+0.12	−1.6	51	76	+25	0.16	0.23	+0.07
660	+0.58	−1.5	51	76	+25	0.08	0.12	+0.04
640	+0.84	−1.5	53	78	+25	0.05	0.07	+0.02
620	+1.26	−1.7	85	110	+25	0.05	0.06	+0.01
Total	−0.77	−4.1	52	77	+25	3.22	4.57	+1.35

Notes: This table reports outcomes under a uniform 25 basis point increase in upfront g-fees across all FICO-LTV cells, in a market recalibrated to an average days-on-market of 64 days (the 2019 pre-pandemic average) by raising the seller entry intercept γ_0^S to −3.111 while holding γ_1^S at its calibrated value. Markups are LLPA minus model-implied cost, in basis points. Profits are markup $\times$ GSE origination volume, scaled to \$616B total GSE originations. Old markups and profits correspond to the post-May 2023 LLPA schedule.

Table A.17: Effect of Uniform G-Fee Increase (+25 bps) on Home Purchases by Income in a Higher-DOM Market (DOM = 64), Calibrated γ_0^S, γ_1^S

Income Group	Δ Purchases (percentage point)
LMI	−0.69
g_1 (low)	−0.71
g_2	−0.85
g_3	−0.95
g_4 (high)	−0.99
Total	−0.77

Notes: Change in total home purchases under a uniform 25 basis point g-fee increase relative to the post-May 2023 LLPA schedule, by income group, in a market recalibrated to an average days-on-market of 64 days under logit seller entry.

Table A.18: Effect of May 2023 G-Fee Change on GSE Markups and Profits, Calibrated γ_0^S, γ_1^S

FICO	Δ Purch.%	Δ GSE%	GSE Markup (bps)			GSE Profit (\$B)		$\Delta\pi$ (\$B)
			Old	New	Δ	Old	New	
780	+3.71	+5.6	66	56	−10	1.28	1.14	−0.14
760	−2.06	−2.9	41	51	+10	0.47	0.56	+0.09
740	−4.37	−6.3	24	52	+28	0.23	0.47	+0.24
720	−4.05	−6.1	15	46	+31	0.12	0.33	+0.21
700	−0.64	−1.0	38	42	+4	0.22	0.23	+0.02
680	−0.61	−1.2	46	51	+6	0.18	0.20	+0.02
660	+3.92	+14.3	111	50	−61	0.21	0.11	−0.10
640	+3.14	+18.4	128	54	−73	0.13	0.06	−0.07
620	+1.12	+11.8	135	86	−49	0.08	0.06	−0.03
Total	+0.01	+0.1	47	52	+4	2.92	3.17	+0.24

Notes: This table reports the effect of the 2023 LLPA reform. Δ Purch.% is the change in total home purchases across all channels. Δ GSE% is the change in GSE origination volume. GSE markups are actual LLPA minus model-implied cost, in basis points. Profits are markup $\times$ GSE origination volume, scaled to \$616B total GSE originations. Seller entry follows the logit specification in Equation (6), with (γ_0^S, γ_1^S) calibrated so the model's aggregate home purchase response to a uniform 100 bps rate increase matches [Hacamo \(Forthcoming\)](#).

Table A.19: Effect of May 2023 g-fee change on Home Purchases by Income, Calibrated γ_0^S, γ_1^S

Income Group	Δ Purchases (percentage point)
LMI	+0.07
g_1 (low)	+0.03
g_2	−0.06
g_3	−0.08
g_4 (high)	−0.16
Total	+0.01

Notes: Change in total home purchases across all channels following the 2023 LLPA reform, by income group, under logit seller entry calibrated to the aggregate interest elasticity of [Hacamo \(Forthcoming\)](#).